

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(KMIT)

Narayanaguda, Hyderabad – 500029

(Affiliated to AICTE and JNTUH)

(Autonomous)



ENGINEERING WORKSHOP LAB MANUAL

(For I B.Tech)

(COMMON TO ALL BRANCHES)

DEPARTMENT OF HUMANITIES & SCIENCES



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(Autonomous)

(Approved by AICTE & Govt of T.S and Affiliated to JNTUH)

3-5-1026, Narayanaguda, Hyderabad-29. Ph: 040-23261407

ENGINEERING WORKSHOP(EWS)
RUBRIC MATRIX GRADING

S.No	Criteria	5	4	3	2	1
1	Safety measures followed	Y				
2	Written part	Y	Y			
3	Job work	Y	Y	Y		
4	Job Accuracy	Y	Y	Y	Y	
5	Result	Y	Y	Y	Y	Y



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY (Autonomous)

(Approved by AICTE & Govt of T.S and Affiliated to JNTUH)

3-5-1026, Narayanaguda, Hyderabad-29. Ph: 040-23261407

ENGINEERING WORKSHOP (EWS)

LIST OF TRADES & EXPERIMENTS (FOR EXERCISES)

1. **CARPENTRY**
2. **FITTING**
3. **TIN SMITHY**
4. **HOUSE WIRING**
5. **WELDING**
6. **3D PRINTING**
7. **CNC ENGRAVING**

LIST OF TRADES (FOR DEMONSTRATION)

1. **POWER TOOLS**
2. **PLUMBING**

CARPENTRY

INTRODUCTION:

Wood work or carpentry deals with making joints for a variety of applications like door frames, cabinet making furniture, packing etc.,

Timber:-

Timber is a name obtained from well grown plants or trees. The timber must cut in such a way that the grains run parallel to the length. The common defects in timber are knots, wet rot, dry rot etc.,

Market sizes of timber:-Timber is sold in market in various standard shapes and sizes. They are:-

Log: The trunk of a tree, which is free from branches.

Balk:-The log sawn to have roughly square cross section.

Post:-A timber piece, round or square in cross section with more than 275 mm in width, 50 to 150 mm in thickness and 2.5 to 6.5 mts length.

Board:-A sawn timber piece, below 175 mm in width and 30 mm to 50 mm in thickness.

Reapers:-Sawn timber pieces of assorted and nonstandard sizes, which don't conform to the above shapes.

WORK HOLDING TOOLS:

Carpentry vice:-

It is a work holding device. When handle vice is turned in a clockwise direction, the sliding jar forces the work against the fixed sawn. The greater the force applied to the handle, the tighter to the work held.

Bar clamp:-

It is a rectangular (or) square block with V-groove on one or both sides opposite to each other. It holds cylindrical work pieces.

C-Clamp:-This is used to hold work against an angle plate or V-block.

MARKING AND MEASURING TOOLS: Try square:-

It is used for marking and testing the squareness of planed surfaces. It consists of a steel blade, fitted in a cast iron stock. It is also used for flatness. The size of a try square used for varies from 150 mm to 300 mm, according to the length of the blade. It is less accurate when compared to the try square used in fitting shop.

Fig: steel tape

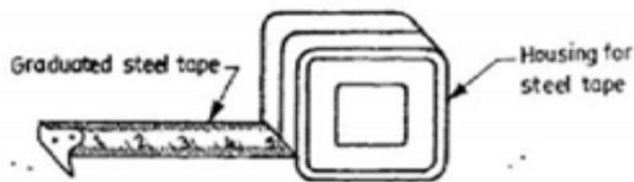


fig: Try square

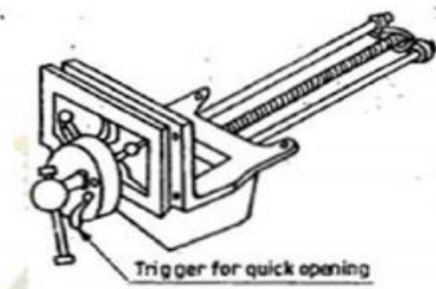
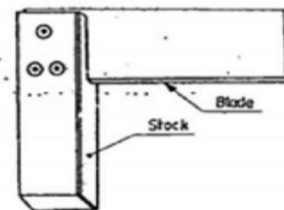


Fig: carpenter vice

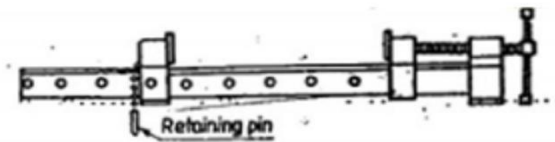


Fig: Bar clamp

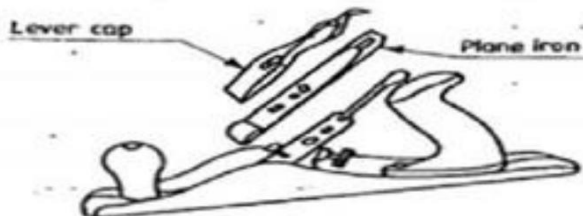


Fig: metal jack plane

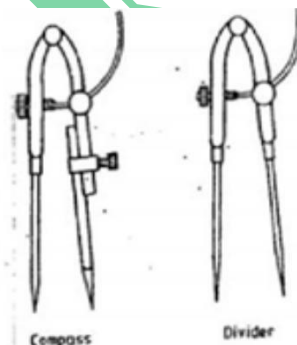


Fig: compass and divider

Marking gauge:-

It is a tool used to mark lines parallel to the edges of wooden pieces. It consists of a square wooden stem with a riding wooden stock on it. A marking pin, made of steel is fitted on the stem. A mortise gauge consists of two pins. In these it is possible to adjust the distance between the pins, to draw two parallel lines on the stock.

Compass and dividers:-

This is used for marking circles, arcs, laying out perpendicular lines on the planed surface of the wood.

CUTTING TOOLS:

Hack saw:-

It is used to cross cut the grains of the stock. The teeth are so set that the saw kerfs will be wider than the blade thickness. Hard blades are used to cut hard metals. Flexible blades are having the teeth of hardened and rest of the blade is soft and flexible.

Chisels:-

These are used for removing surplus wood. Chisels are annealed, hardened and tempered to produce a tough shank and a hard cutting edge.

Rip saw:-

It is used for cutting the stock along the grains. The cutting edge of this saw makes a sleeper angle about 60° whereas that saw makes an angle of 45° with the surface of the stock.

Tenon saw:-

It is used for cutting tenons and in fine cabinet works. The blade of this saw is very thin and so it is used stiffed with back strip. Hence, this is sometimes called back saw. The teeth shapes similar to cross cut saw.

DRILLING AND BORING TOOLS:

Auger bit:-

It is the most common tool used for boring holes with hand pressure.

Gimlet:-

This is a hand tool used for boring holes with hand pressure.

Hand drill:-

Carpenters brace is used to make relatively large size holes, whereas hand drill is used for drilling small holes. A straight shank drill is used with these tools. It is small light in weight and may be conveniently used than the brace. The drill is clamped in the chuck.

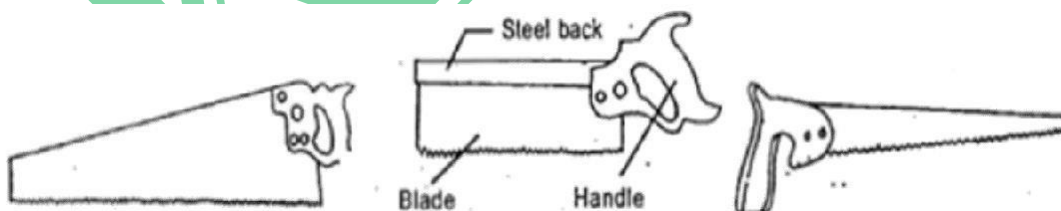


Fig: cross cut saw

Fig: Tenon saw

Fig: compass saw

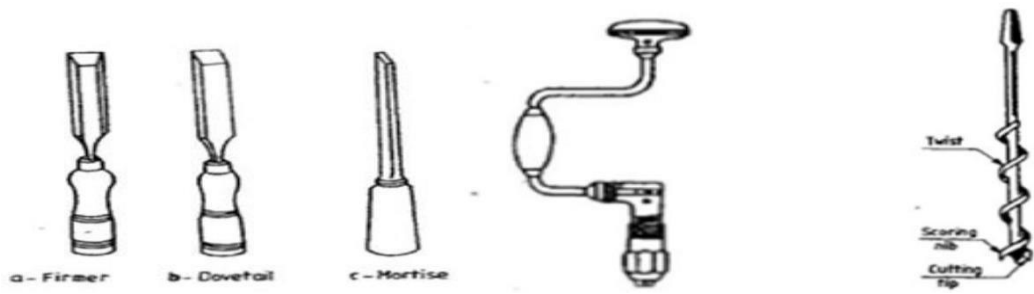


Fig: Chisels

Fig: Carpenter's brace

Fig: Auger bit

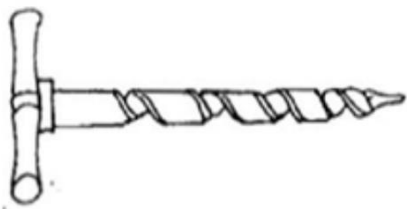


Fig: Gimlet



Fig: wood rasp file

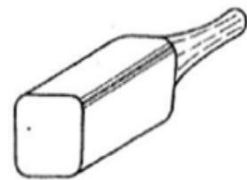


Fig: Mallet

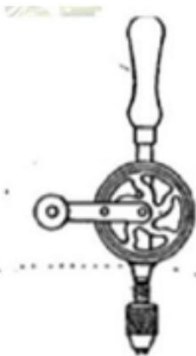


Fig: Hand drill

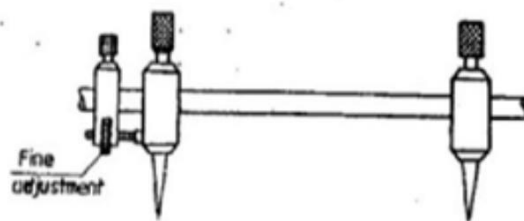


Fig: Trammel

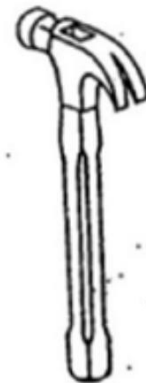


Fig: Claw hammer

MISCELLANEOUS TOOLS:

Ball peen hammer:-

It has a flat face, which is used for general work and a ball end is used for riveting.

Mallet:-

It is used to drive the chisel, when considerable force is to be applied, steel hammer should not be used for these purpose, as it may damage the chisel handle. Further, for better to apply a series of light taps with the mallet rather than a heavy single blow.

Claw hammer:-

It is a striking flat at one end and the claw at the others. The face issued to drive nails into wood and for other striking purpose and the claw for extracting nails out of wood.

Pinches:-

It is made of steel with a hinged and is used for pulling out small nails from wood.

Wood rasp file:-

It is a finishing tool used to make the wood smooth, remove sharp edge finishing fillets and other interior surfaces. Sharp cutting teeth are provided on its surface for the purpose. This file is exclusively used in wood work.

Exercise -1

T- LAP JOINT

Aim: To make the T- Lap joint the required dimensions from the given work piece.

Material Required: Soft wood of size 300 x 50 x 50 mm.

Tools Required

1. Jackplane
2. Carpentry vice
3. Try square
4. Marking gauge
5. Steel rule
6. Tenon saw
7. Rip saw
8. Firmer chisel
9. Mallet

Procedure :

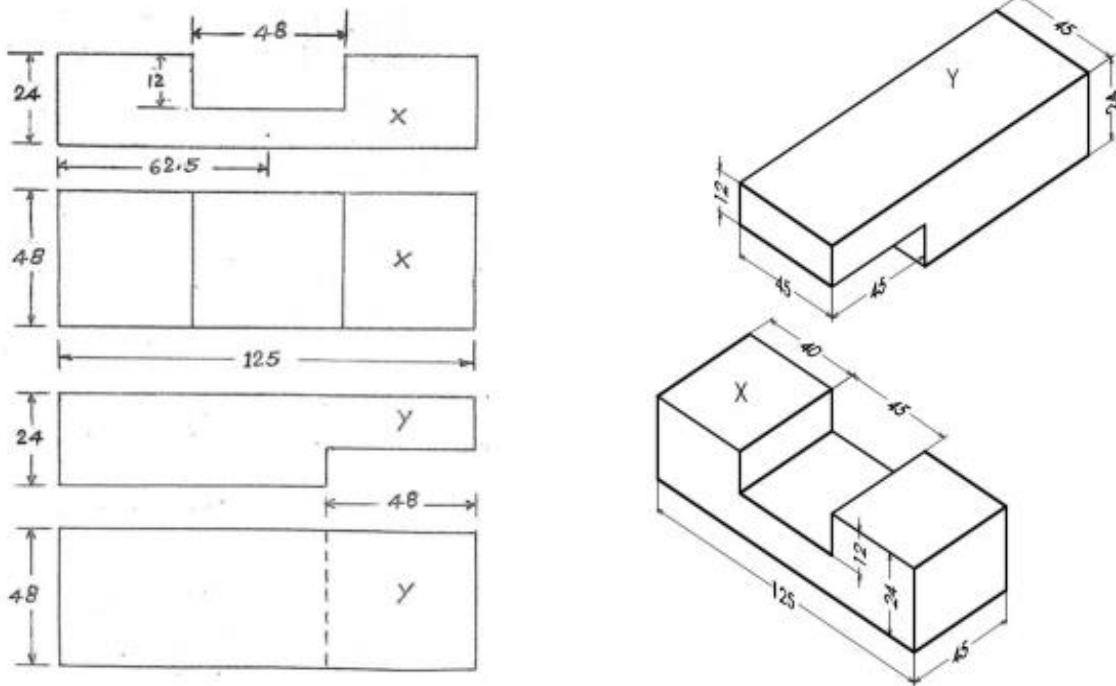
1. The given work piece is firmly clamped in the carpentry vice and any two adjacent surfaces are planed to get right angles using the jack plane.
2. Using the try square, the right angles of planned faces are checked.
3. Now the other two surfaces are planned to get smooth surface.
4. The work piece is cut into two pieces by using the rip saw.
5. Using the steel rule and marking gauge, marking is done for T-joint on the two halves.
6. In one half, the unwanted portions of wood are removed by using the tenon saw and firmer chisel. The same procedure is done for the other half of work piece.
7. Using the jack plane, the other two faces of work piece is planned to the required size.
8. The finished two pieces are assembled to get together to form the T-joint.
9. Finally, the finished job is checked for required size and shape using the steel rule and try square.

Precautions:

1. Wood should be free from moisture
2. Marking is done without parallax error
3. Care should be taken while chiseling
4. Matching of X and Y pieces should be tight.

Result:

Thus the required T- Lap Joint is obtained.



All dimensions are in mm

Exercise No -2

DOVETAIL LAP JOINT

AIM: To make a Dovetail Lap joint experiment from the given wood piece.

MATERIAL REQUIRED: Teak wood piece of size 50 X 30 X 125 mm – 2 pieces

TOOLS REQUIRED:

1. Governor Setup,
2. Carpenter's vice,
3. Steel Rule,
4. Jack plane,
5. Try-square,
6. Marking Gauge,
7. Firmer Chisel,
8. Cross-cut Saw,
9. Tenon- Saw,
10. Scribe,
11. Wooden Mallet and
12. Wood rasp File.

PROCEDURE:

1. The given wood pieces are checked to confirm size as per sketch given.
2. The wood piece is firmly fixed in the carpenter's vice and any two adjacent faces are planned by jack plane and the two other faces are checked for the squareness using the try-square.
3. Making gauge is set and lines are drawn at 24 and 48mm, to mark the thickness and width of the model respectively.
4. The portions to be removed in both male and female pieces can be cut by using the cross cut saw, followed by chiseling.

5. A fine finishing is given by using wood rasp file.
6. The ends of both parts are chiselled and filed by Wood rasp file.
7. A fine finishing is given to the parts, if required so that, proper fitting is obtained.
8. The parts are fitted to obtain a slightly tight joint

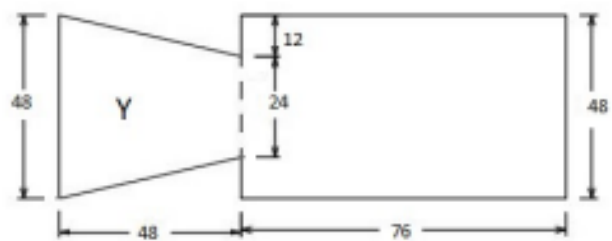
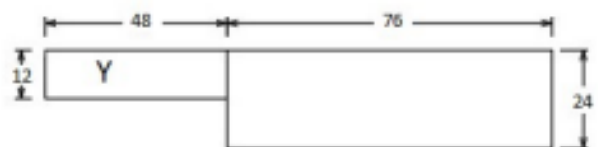
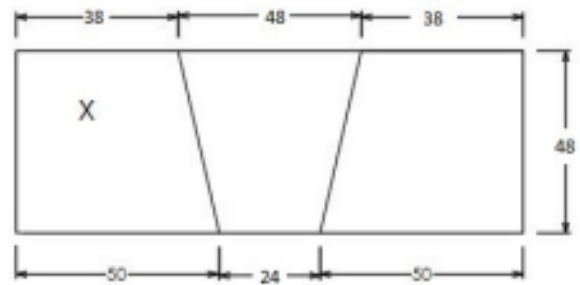
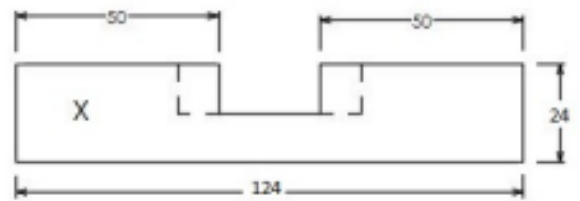
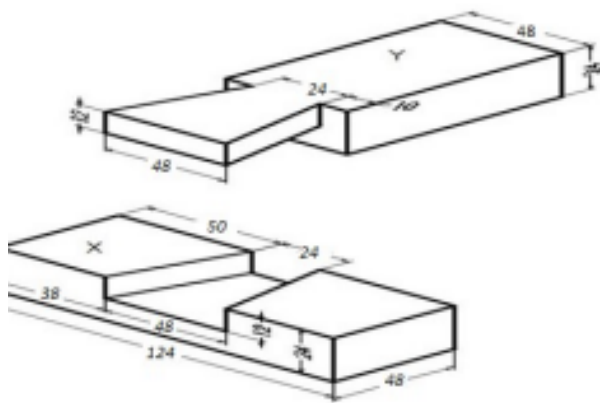
Precautions:

1. Do not use firmer chisel without handle.
2. Do not put finger on cutting edge of chisel and Saw.

RESULT & CONCLUSION:

The Dovetail Joint is thus made by following the above sequence of operations

DIAGRAM:



All dimensions are in mm

Viva Questions

- Q1: What is the sequence of operations in carpentry?
- Q2: Classify the tools used in carpentry?
- Q3: Name some measuring tools?
- Q4: Name some marking tools?
- Q5: Name some cutting tools?
- Q6: What is the use of mortise gauge?
- Q7: What is the use of wooden jackplane?
- Q8: What is the difference between saws and chisels?
- Q9: What are the various types of joints?
- Q10: Which joint is the strongest? Justify your answer.
- Q11: What is the difference between marking gauge and mortise gauge?
- Q12: What is the use of a claw hammer?
- Q13: What safety precautions should be taken in carpentry?
- Q14: Why is butt joint not performed in carpentry?
- Q15: How does a rip saw differ from a crosscut saw?.

FITTING

INTRODUCTION:

Machine tools are capable of producing work at a faster rate, but there are occasions when components are processed at a bench. Sometimes it becomes necessary to replace or repair a component that must fit accurately with one another or reassemble. This involves a certain amount of hand fitting. The assembly machine tools, jigs, gauges etc., involves certain amount of bench work.

FITTING TOOLS:

Holding tools:-

1. Bench vice
2. V-block with clamp
3. C-clamp

Bench vice:-

It is a work holding device, when vice handle is turned in a clockwise direction the sliding jaw forces the work against the fixed jaw, the greater the force applied to the handle, the tighter is the work held.

V-block with clamp:-

It is a rectangular (or) square block with v-groove on one or both sides, opposite to each other. It holds cylindrical work pieces.

C-clamp:-

This is used to hold work against an angle plate or v-block.

MARKING AND MEASURING TOOLS:

- | | |
|-----------------------------|---------------------|
| 1. Surface plate | 6. Odd leg caliper |
| 2. Try square | 7. Divider |
| 3. Angle plate | 8. Calipers |
| 4. Scriber | 9. Dot punch |
| 5. Universal scribing block | 10. Vernier caliper |

Surface plate:-

It is used for testing flatness of work piece, for marking out small works.

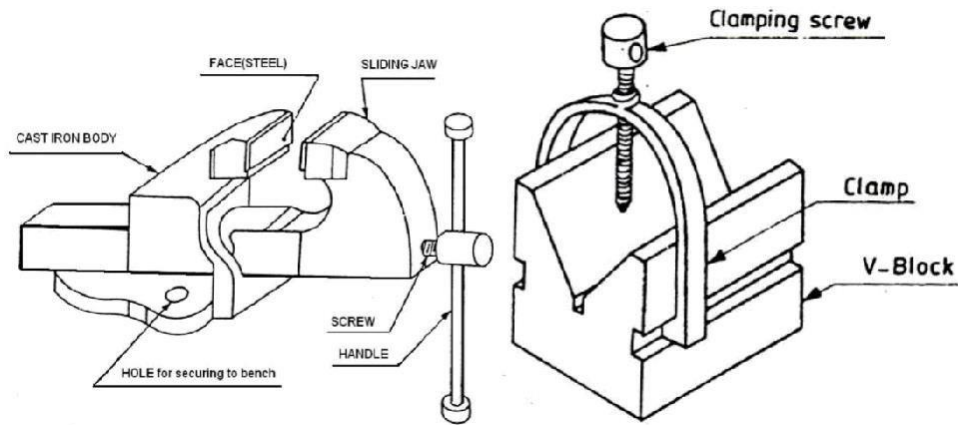


Fig: 1 Bench vise

Fig: 2 V-Block

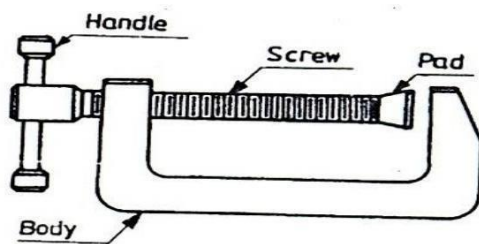


Fig: 3 C – Clamp

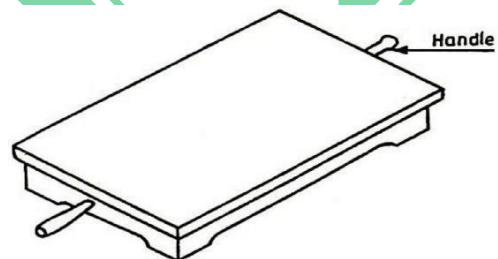


Fig: 4 Surface plate

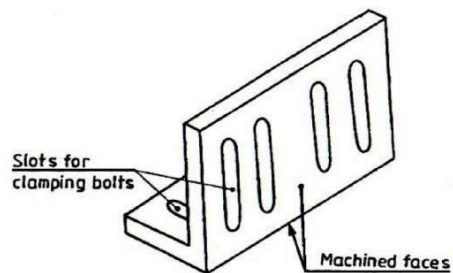


Fig: 5 Angle plate

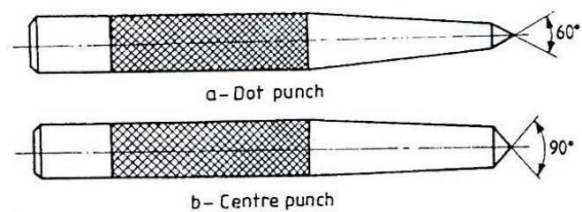


Fig: 6 Dot punch

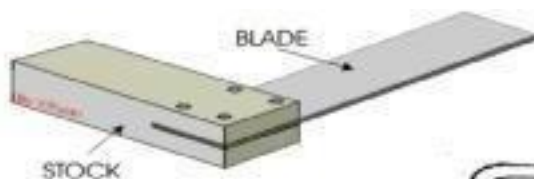


Fig: 6 try square

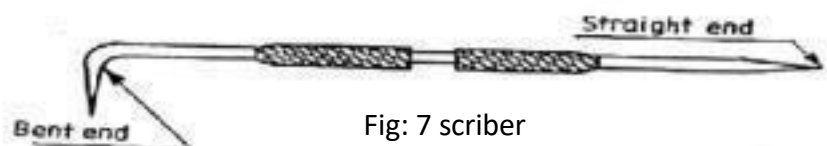


Fig: 7 scribe

Combination cutting pliers: -

This is made of tool steel and is used for cutting as well as for ripping work.

Taps and die holders: -

Tap and wrenches are used for cutting internal threads in a drilled hole.

Dies and die holders:-

They are used for making external threads. Dies are made either solid (or) split type.

TYPES OF FILES: Hand file:-

It is a rectangular in section tapered in thickness but parallel in width.

Flat file:-

Rectangular in section and tapered for $1/3^{\text{rd}}$ length in width and thickness.

Square file:-

Square in section and tapered for $1/3^{\text{rd}}$ length on all sides.

Half round file:-

It has one flat face, connecting by a curved (surface) face & tapered for $1/3^{\text{rd}}$ length.

Round file:-

Circular in cross section and tapered for $1/3^{\text{rd}}$ length, it has double cut teeth.

MISCELLANEOUS TOOLS:**Ball peen hammer:-**

It has a flat face, which is used for general work and a ball end is used for riveting.

Screw driver:-

It is designed to turn the screws. The blade is made of steel and is available in different lengths and diameters.

Spanners:-

It is a tool for turning nuts and bolts. It is usually made of forged steel.

FITTING OPERATIONS

Chipping:-

Removing metal with a chisel is called chipping and is normally used where machining is not possible.

Fitting:-

1. Pinning of files:-

Soft metals cause this; the pins are removed with a file card.

2. Checking flatness and square ness:-

To check flatness across thickness of plate.

Marking and measuring:

Measurements are taken either from a center line, for visibility of the non-ferrous metals and oxide coated steels are used.



Fig: 8 odd leg clamp and divider

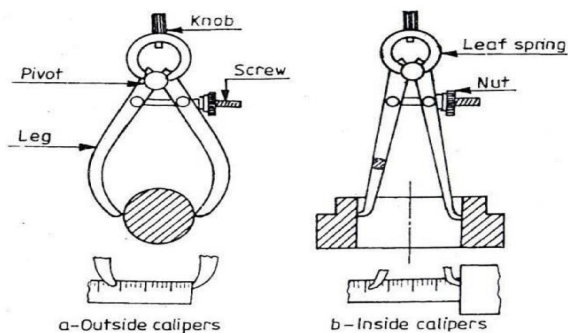


Fig: 9 calipers

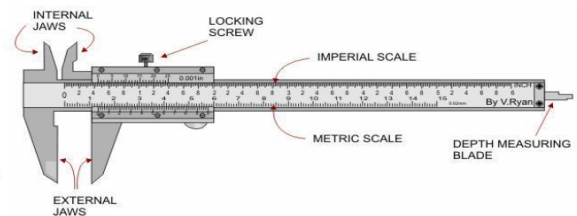


Fig: 10 Vernier caliper

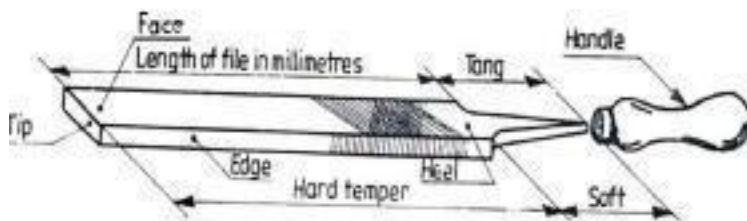


Fig: 11 Parts of hand file

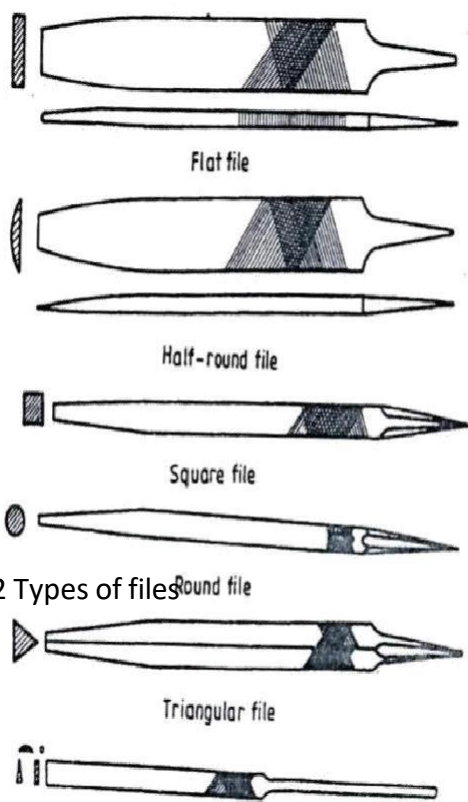


Fig: 12 Types of files

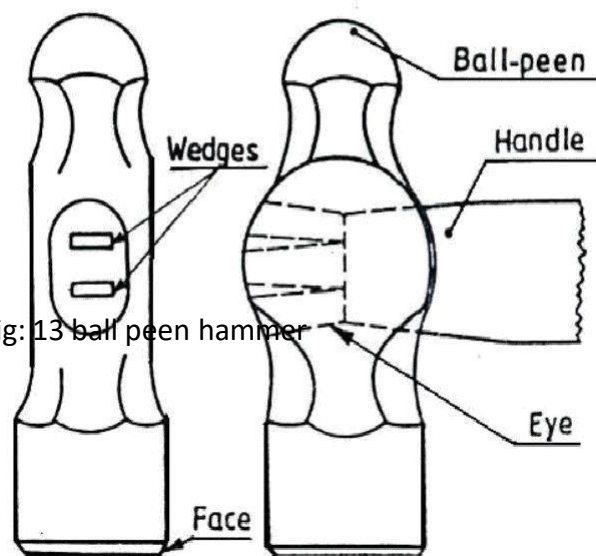


Fig: 13 ball peen hammer

EXERCISE NO 01

V- FITTING

AIM: To make V-fit from the given two M.S. pieces.

MATERIAL REQUIRED: M.S. Piece of size: 50 X 50 X 5 mm – 2 Pieces

TOOLS REQUIRED:

1. Bench Vice,
2. Steel Rule,
3. Try Square,
4. Ball-Peen Hammer,
5. Scriber, Dot Punch,
6. Set of Files,
7. Surface Plate,
8. Jenny Caliper,
9. Hacksaw with Blade and
10. Flat Chisel.

PROCEDURE:

1. The burrs in the pieces are removed and the dimensions are checked with the steel rule.
2. The pieces are clamped one after the other and the outer mating edges are filed and checked for their flatness with the help of the try-square
3. The side edges of the two pieces are filed such that, they are at right angle to each other and widths are exactly 48 mm.
4. Chalk is then applied on the surfaces of the two pieces.
5. The given dimensions of the “V” fitting are marked, by using the Jenny caliper, steel rule and surface plate.

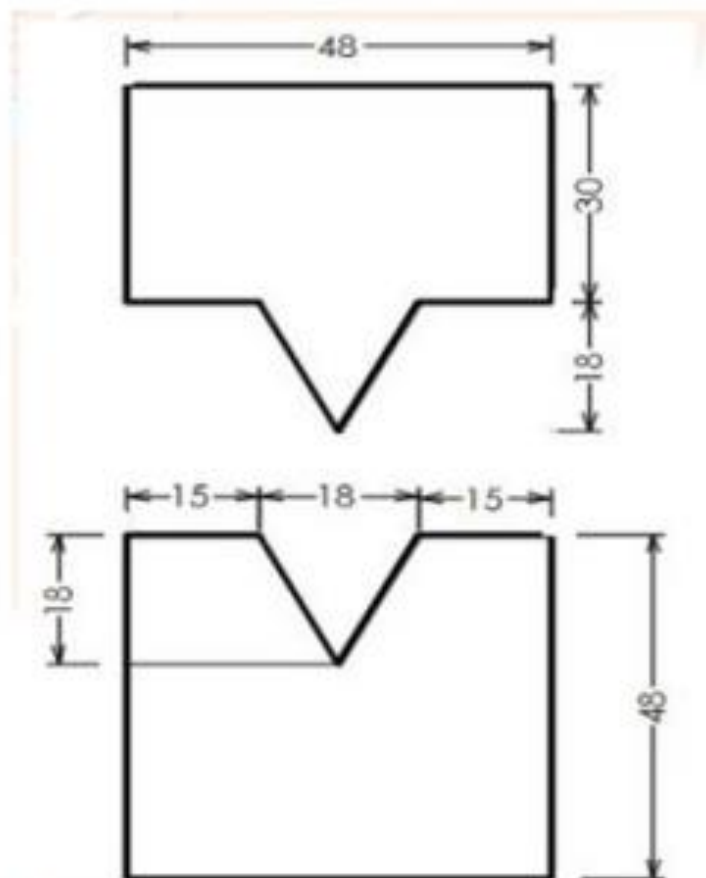
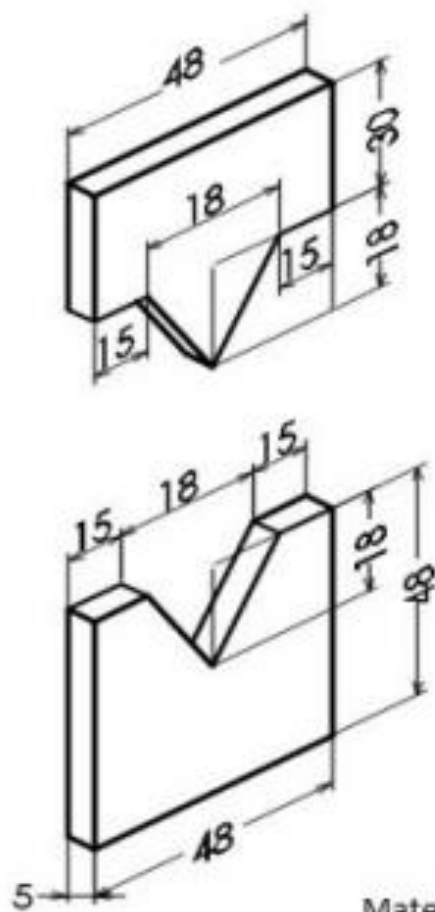
6. Using the dot punch, dots are punched along the above scribed lines.
7. Using the hack saw, the unwanted portions are removed.
8. Using the flat chisel, the unwanted material in the piece Y is removed.
9. The cut edges are filed by the triangular file.
10. The corners of the stepped surfaces are filed by using square or triangular file to get the sharp corners.
11. The pieces (X and Y) are fitted together and the mating is checked for the correctness of the fit. Any defects noticed, are rectified by filing with a smooth file. Note: Care is taken to see that the marking dots are not crossed, which is indicated by the half of the punch dots left on the pieces.

Safety precautions: -

1. Care is taken to see that the marking dots are not crossed, which is indicated by the half of the punch dots left on the pieces.
2. Apply pressure in forward direction during hack sawing.
3. Don't rub steel rule on the job.
4. Fix blade in hack saw frame with correct tension.
5. During hack sawing the coolant like water or lubricating oil is to be used.
6. Use precision instruments like vernier calipers and vernier height gauge carefully.
7. Files are to be cleaned properly after using.

Result: - V- fit is made as per the required dimensions.

DIAGRAM:



Material: Mild Steel

All Dimensions are in mm

EXERCISE NO: 2

SQUARE FIT

Aim: To make square fitting from the given two M.S pieces.

Material: Two MS FLAT of size 50x50x5mm

Tools required:

150mm try-square, 250gm ball-peen hammer, dot punch, scribe, chisel, 300mm hacksaw frame, 18TPI hacksaw blade, 250mm rough and smooth hand files, 6mm rough and smooth square files, Bench vice and steel rule.

Sequence of operations:

1. The burs in given materials are removed and the dimensions are checked for 50x50x5mm with steel rule.
2. The pieces are clamped one after the other and outer mating edges are filed and checked for their flatness, with the help of try-square.
3. The side edges of the two pieces are filed such that, they at right angles to each other and widths are exactly 48mm.
4. Wet chalk is applied on surfaces of the two pieces.
 - The given dimensions of the Stepped fitting are marked, by using jenny caliper, steel rule scribe, and the surface plate.
 - The portion to be removed is then marked.

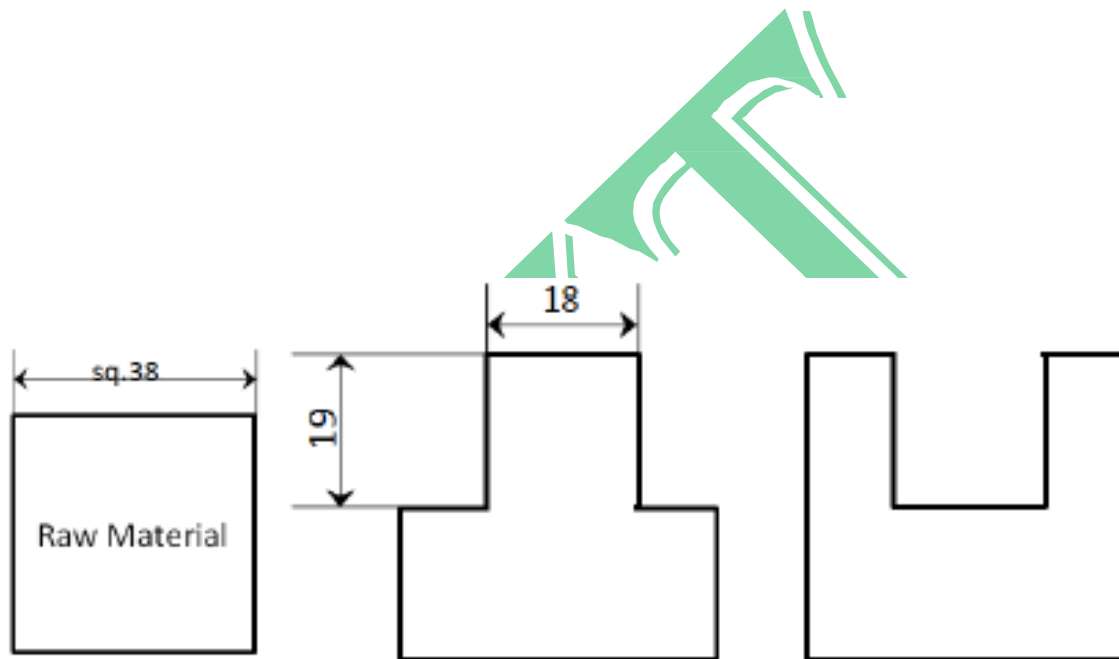
Using dot punch, dots are punched along the above scribed lines. Using the hacksaw, the unwanted portions are removed.

Now the portions are filed and burrs are removed by filing on the surfaces of fitted job.

Precautions:

1. Never remove chips with hand use a wire brush.
2. Working tool should not be kept at the edge of table.
3. While sawing secure work rigidity.
- 4.

Result: The Square fit is thus made by following the above sequences of operations.



VIVA QUESTIONS:

1. What is meant by fitting ?
2. Name the commonly used material for preparing fitting models ?
3. Specify the composition of M.S.
4. What are commonly used tools in fitting shop ?
5. What is a vice ? Name the different types of vices ?
6. What is file ? Name the different types of files ?
7. Define filing ? Name the materials used in the manufactures of files.
8. What is meant by single cut & double out files.
10. Name different types of Hammers ?
11. What are hard & soft Hammers ?
12. What is a chisel ? Name the different types of chisel ?
13. Define chipping ?
14. Differentiate between cold chisel & hot chisel ?

TIN-SMITHY

Sheet metal work is applied to the working of thin metallic sheets with hand tools and sheet metal machines. Many important engineering articles made up of sheet metal and their application in air-conditioning ducts, aircraft industry, agriculture implements, decorative articles and household articles. For effectively working in sheet metal one should have knowledge of hand tools, sheet metal machines, properties of metals and thorough knowledge of projective geometry i.e. development of surfaces.

Types of sheet Metal

1. Ferrous Sheet

- i) **Mild Steel sheets** – These are black iron sheets, susceptible to rust and corrosion, mostly used in water tanks and fabrication works.
- ii) **Galvanized Iron (GI Sheet)** – It is soft steel sheet coated with zinc, sheets have corrosion resistance due to zinc coating, used for making air-conditioning ducts, roofs, boxes, buckets, coolers etc.
- iii) **Stainless steel sheets** – It is an alloy of high grade steel with chromium, nickel, phosphorous and manganese. It is used in household goods, food processing plants.
- iv) **Tin Plate** – steel coated with tin is called Tin steel. It is used for making food containers.

2. Non-ferrous sheets

- i) **Aluminium Sheets** – It is two and half times lighter than iron but lacks in tensile strength. Small percentage of other elements like copper, manganese and silicon is added to make it suitable for production in air-craft industry and other industrial goods. It is also called aluminium alloy sheets.
- ii) **Copper and Brass sheets** – These are non-ferrous sheets used in electrical and various other industrial and household articles.

Measurement of Thickness of Sheets

Thickness of sheet is generally indicated by gauge number, which is obtained by actually measuring the sheet thickness with a sheet gauge or wire gauge. Each slot in the standard wire gauge is numbered, a number which represents gauge number such as 20 SWG. The more the wire gauge number, less is the thickness of sheet.

Tools used in Sheet Metal

1. Common Hand Tools – Hammer, Files, Hacksaw, Chisels, Punches, Steel rule, Try square, Scriber.

Straight Snips – It is used to cut small thickness of sheets along straight line. To cut thickness of greater thickness shearing machine.

2. Bent Snip – The cutting blades are curved from cutting edge. It is used to cut discs and round articles from sheets.
3. Trammel Points – It is used for drawing large circles and arcs. It is two straight, removable legs tapered to the middle point and mounted on separate holders.
4. Mallet – Mallet is made up of good quality of hard wood; this is used whenever light force is required. Use of mallet does not spoil the surface of sheet; it is used for smothering of sheet.
5. Groover – It is a tool used to make a locked grooves seam in sheet metal joints.

Supporting Tools

Stakes – Stakes are used to support sheets in bending, seaming, forming, riveting, punching etc. Some commonly used stakes are:

1. Half Moon Stake – It is used for working the edges of discs.
2. Hatchet Stake – It is used for forming, binding and seaming the edges.
3. Bick Iron – It is used for forming long tapered cylindrical components.
4. Funnel Stake - It is used for forming conical components.
5. Pipe Stakes – It is used for forming pipes and hollow cylindrical surfaces.

Sheet Metal Joints and Strengthening Processes

1. Lap Joint – It can be prepared by means of soldering or riveting.
2. Seam Joint – When two or more sheets are folded and fastened together is called seam joint. There are two types of seam joints.
 - i) Single Seam Joint
 - ii) Double Seam Joint

3. Groove Seam Joint – In this two single edges are hooked together and flattened with a small mallet to make them tight, seam is then grooved with a hammer and hand groover.
4. Wired Edge – It is one of the methods of strengthening the thin metal by turning over the edge on a wire in it.
5. Hinged Joint – It is used for easy movement of opening or closing doors, window etc.
6. Cap Joint
7. Hem Joint

Sheet Metal Operations:

1. Measuring and Marking – Sizes are marked on large sheet to cut the latter into small pieces.
2. Development of Surface (Laying Out) – It means the operation of scribing the development of surface of the component on the sheet together with the added allowance for overlapping, bending, hammering etc.
3. Cutting and shearing – The term shearing stands for cutting of sheet metal by two parallel cutting edges moving in opposite direction.
4. Hand Forming – It stands for shaping, bending of sheet metal in three dimensions in order to give the desired shape and size of final product.
5. Nibbling – It is a process of continuous cutting along a contour which may be of straight or irregular profile.
6. Piercing and Blanking – Piercing is basically a hole punching operation while blanking is an operation of cutting out a blank.
7. Edge Forming or Wiring – Edges of sheet metal products are folded to provide stiffness to the products and to ensure safety of hand due to sharp edges.
8. Joint Making – Sheet metal parts can be joined by folded joints, riveting, welding, brazing, soldering, self tapping screws, screwed fastening, and by adhesives.
9. Bending – Bend in sheet metal is to be bent at different angles to shape it to required form.
10. Circle Cutting – It is an operation of cutting circular blanks or curved contours with the help of circular cutting machines.

11. Hollowing – It is the process whereby a flat sheet metal is beaten up into spherical shape by placing the metal upon a sand bag or hollowing block, beating with hollowing hammer, starting from boundaries towards center.
12. Raising – It is the process of hammering the metal from oxide to form a hollow article, working around from center towards edge. Turned over Edge – It is the method of strengthening the thin metal at edge. The edges are turned with some radius.
13. Swaging – This is also a method of strengthening thin sheet metal by making impressions in the bodies. It is done by machine or by hand.

Sheet Metal Machines

1. Shearing Machine – This is used to cut sheets. They are of two types:
 - i) Hand operated – Used for cutting thin sheets.
 - ii) Power operated – Used for cutting thick sheets.
2. Folding Machine – It is used for folding the sheet edges to form the joint.
3. Swaging Machine – It is provided with different types of contours on sheets to give strength to thin sheets.
4. Rolling Machine – It is used for shaping metal sheets into cylindrical objects, this machine consists of three rollers that can be adjusted for different radii.
5. Circular cutting machine – It is used for cutting circular discs.
6. Bench Shear – This machine is fixed on a bench and used for cutting comparatively thicker sheets.



Square Stake



Hatchet Stake



Half Moon Stake



Funnel Stake



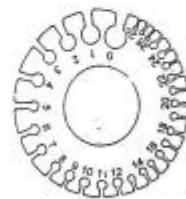
Wooden Mallet



Nylon Mallet



Trammel



Standard Wire Gauge



Bench Shear Machine



Straight Snip



Bent Snip



Bent Scriber



Straight Scriber

Exercise No.-1

RECTANGULAR TRAY

AIM: - To make a rectangular tray.

Tools and materials used: Steel rule, Scriber, Divider, Mallet, Stakes, Try square, snip straight (Tin cutter) Bench shear, file flat smooth, Nylon hammer, Tin sheet piece.

Materials used: - Galvanized iron sheet 28 SWG. **Drawing:** - See Diagrams

Procedure:

1. For developing the surfaces (lying out) draw plan, front view and end view of the required open rectangular box as shown in diagram.
2. Extend all the lines and cut these lines according to the height of the tray (i.e. 30 mm).
3. Addition strips of stocks are given along the edges for single hem allowance (6 mm).
4. All the four corner triangles are cut as shown in diagram for joint making.
5. Perform the operation of cuttings, shearing edges, hand forming, edge forming, joint making and bending in to rectangular tray by using above mentioned sheet metal tools by.
 - I. Square folding along 12, 23, 34, 41 lines.
 - II. Square folding of 15, 26, 37, 48, 2-10, 3-11, 19, 4-12,
 - III. Square fold for Hem 56, 78, 10, 11, 9-12.
6. File the sharp edges with a smooth flat file.

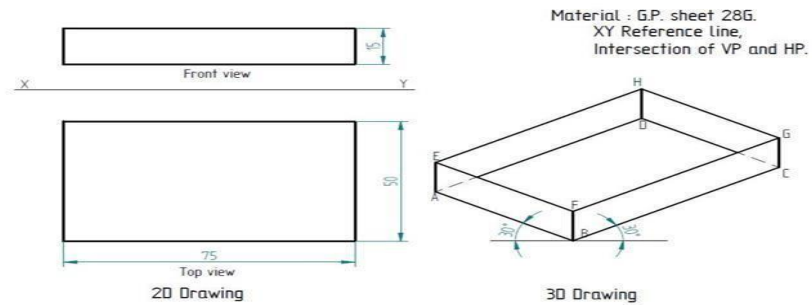
Safety Precautions:

- 1) Take precautions while working on sharp edges of sheets to avoid injury.
- 2) Appropriate cutting tools and machines must be used for cutting tin sheets.
- 3) Avoid using blunt cutting tools.
- 4) Extra allowance must be provided in the sheets while cutting so that finished product is of correct size and finish.

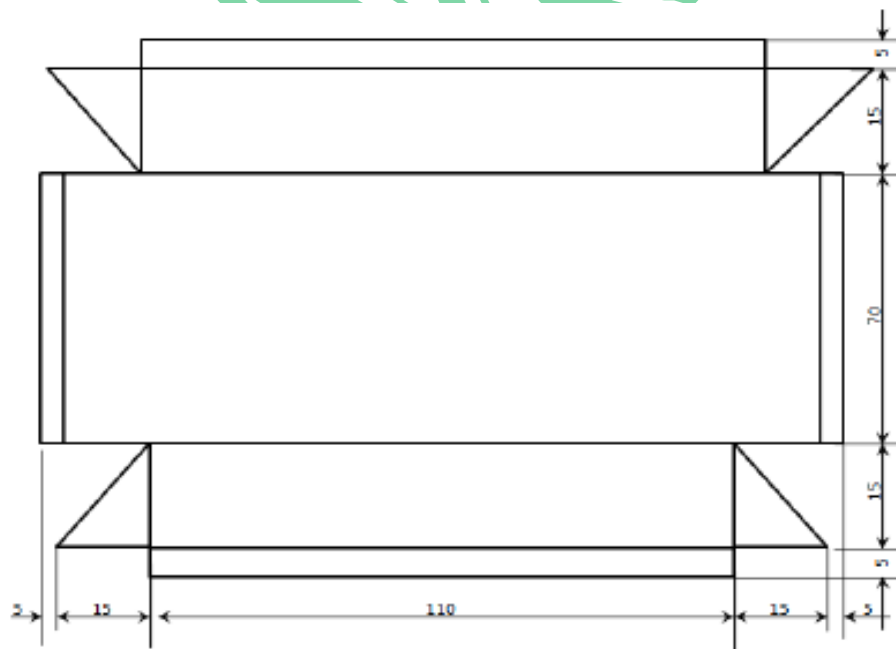
Result:

Rectangular Tray is thus made by following the above sequences of operations.

RECTANGULAR TRAY



All Dimensions are in mm.



Exercise No.-2

SQUARE BOX

AIM: - To make a Square box using G.I Sheet as per the dimension

Tools and Equipment used :-

1. Straight snip,
2. steel rule,
3. scribe,
4. Mallet,
5. Hammer,
6. Stakes,
7. pliers,
8. soldering iron,
9. solder,
10. flux,
11. bench vice,
12. file,
13. spring divider.

Materials used: - Galvanized iron sheet 28 SWG.

Procedure:

1. Draw a lay out as shown in development on drawing sheet.
2. Cut the pattern to shape along the line using a suitable snip.
3. Mark on the G.I Sheet as per the pattern and cut to required shape.
4. Make the hem edge using mallet and stake.

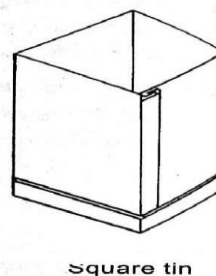
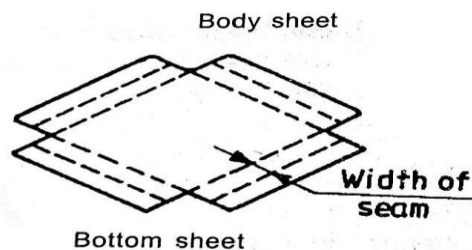
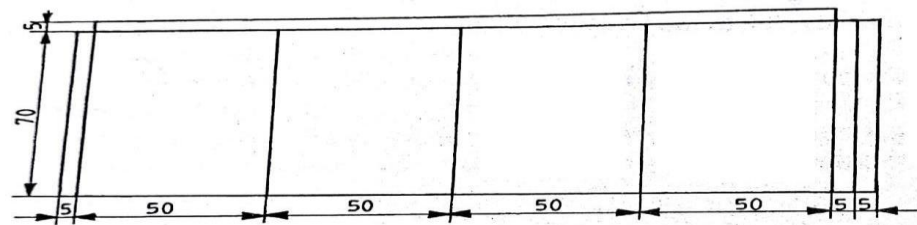
5. Make closed folds on both ends for lock seam joint
6. Make square folds on lines marked A, B, C, D, E, & F
7. Make lock seam joint after joining both the ends.
8. Make a bottom piece from G.I Sheet taking required allowance for double lock seam joints as shown in diagram.
9. Join the bottom piece with square box by double lock seam joint using stakes and mallet.
10. Do the soft soldering operation on the corners of double lock seam joints.
11. File all the sharp corners with file.

Precautions:-

1. Be careful while working on sharp edges of sheets to avoid injury.
2. Do not use blunt cutting edges tool.
3. Appropriate cutting tools and machines must be used for cutting tin sheets.
4. Extra allowance must be provided in the sheets while cutting so that finished Product is of correct size & finish.

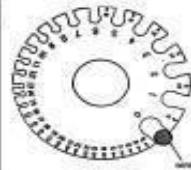
Result :

Square Box is thus made by following the above sequences of operations.

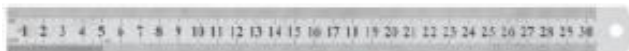


TOOLS USED IN TIN-SMITHY

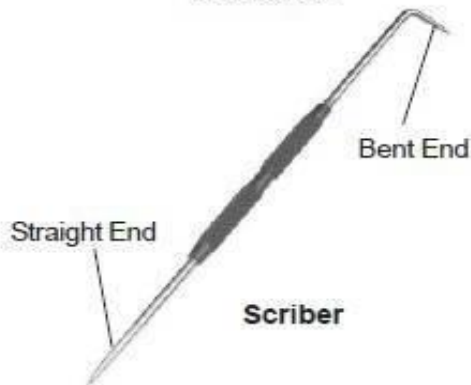
Wire No.	Thickness, mm
10	3.25
12	2.50
14	2.00
16	1.60
20	1.00
22	0.70
24	0.50
26	0.45
30	0.30



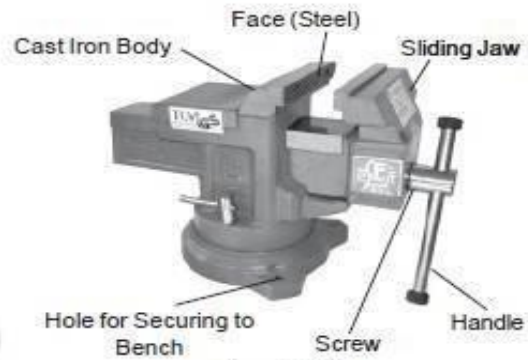
Standard Wire gauge



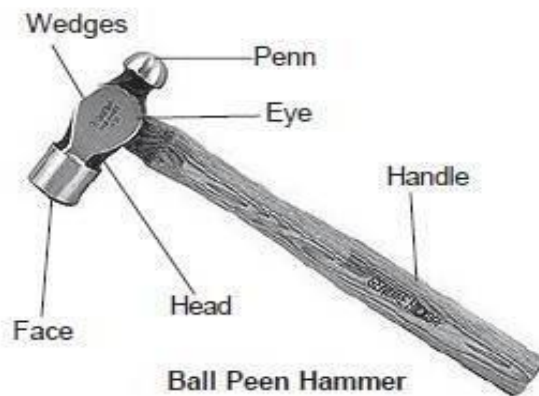
Steel Rule



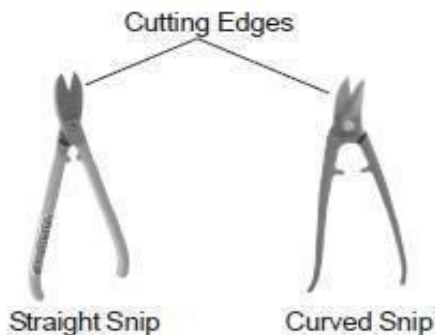
Scriber



Bench Vice



Ball Peen Hammer



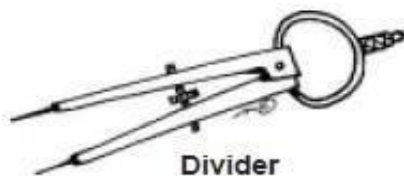
Snips



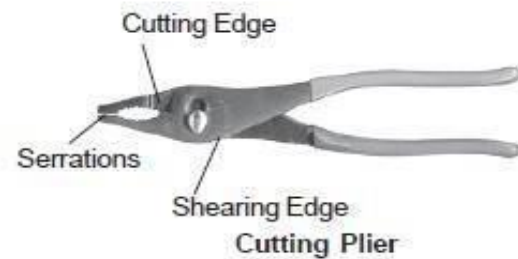
Cross Peen Hammer



Straight Peen Hammer



Divider



Cutting Plier

VIVA QUESTIONS:

1. Describe the process of sheet metal work ?
2. What are the engineering applications of sheet metal work ?
3. Mention the different materials used in sheet metal work ?
4. What are the different methods of joining the sheet metal ?
5. What is a G. I. Sheet ? What is the purpose of wire gauge ?
6. What is a galvanisation ? Why it is required ?
7. Name the different types of stakes used in sheet metal work ?
8. Define soldering ?
9. What are the applications of soldering ?
10. What is a solder ?
11. What kinds of solders are commonly used ?
12. Why lead is used as a solder in soldering ?
13. What is a flux and why it is used ?
14. Which is the most important operation in soldering
15. What is the purpose of using soldering iron ?

HOUSE WIRING

Introduction:

Power is supplied to domestic installations through a phase and neutral, forming a single phase AC 230 v to wire system. For individual establishment power is supplied through three phase two wire system. To give 440V, the neutral is earthed at to the domestic utilities; power is fed to kilo watt meter and then to distributes power along several circuits. It also protects these circuits from over load by safety devices like fuses or circuit breakers.

Elements of house wiring:

- Fuses and circuit breakers
- Electric switch
- Plug
- Socket out let
- Lamp holder
- Main switch
- Incandescent light

Wiring method: A circuit is path along which the electric current flows from negative side of power source to positive side.

Incandescent Light:

Incandescent means glowing at white hot. A lamp actually works like a heating element, except that it gives off light by becoming white hot. Figure 3.9. shows the parts of a bulb. The amount of power it consumes is stamped on the bulb. The higher the wattage, brighter the light. The bulbs have filaments made of tungsten. However, special bulbs are available with inside coating and filled with gas.

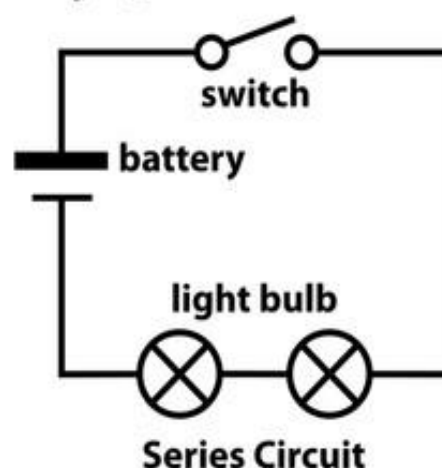
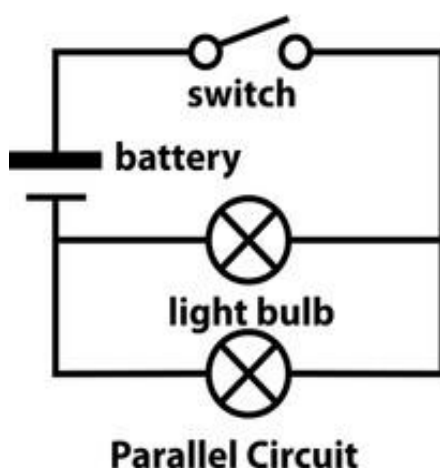
Wires and Wire Sizes:

Wire is defined as a bare or an insulated conductor consisting of one or several strands. An insulated wire consists of a conductor with insulating material made of vulcanized India rubber (VIR) or polyvinylchloride (PVC). The wire may consist of one or several twisted strands. A multi-core conductor consists of several cores insulated from one another and enclosed in a common sheathing (fig. 3.10).

Wire sizes are specified by diameter of the wire, using a standard wire gauge (SWG), which also gives an idea of the current carrying capacity. The specification consists of both the no. of strands.

1. Connection of switches
2. Circuit-connection
3. Power from mains
4. Operating the lamp

Circuit Types



shutterstock.com · 1619901676

Exercise No.1

TWO LAMPS CONTROLLED BY ONE SWITCH IN PARALLEL OR IN SERIES CONNECTION

AIM: To give connection to one Lamp controlled by one Switch.

MATERIAL REQUIRED:

TOOLS REQUIRED:

1. Wooden wiring board,
2. one way switch,
3. wooden round blocks,
4. batten lamp holders,
5. connector,
6. screw driver
7. wires,
8. wire clips,
9. nails,
10. wood screws,
11. poker and bulbs.

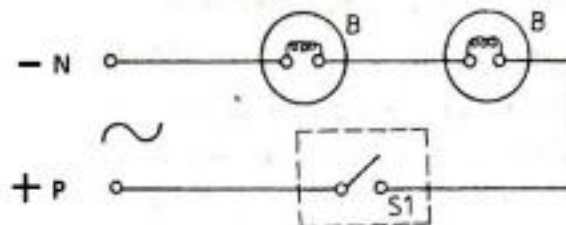
PROCEDURE:

1. The outline of the wiring diagram is marked on the wooden wiring board.
2. Clips are nailed to the board, following the wiring diagram
3. Wires are stretched and clamped with the clips
4. Round blocks (3 No's) are screwed onto the board, as per the diagram.
5. Wires are connected to the holders and switch which are then screwed onto the round blocks.
6. Bulbs are fitted to the holders
7. The wiring connections are then tested, by giving power supply.

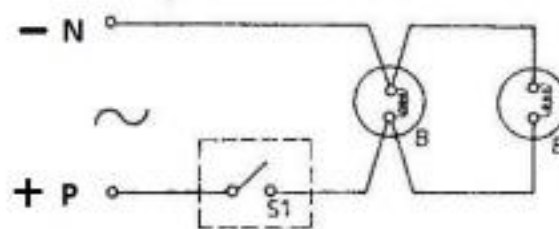
RESULT & CONCLUSION:

The electrical circuit, for two lights controlled by one switch in series is thus made.

DIAGRAM:



Two Lamps controlled by one switch in Series connection



Two Lamps controlled by one switch in Parallel connection

Exercise No.2

ONE LAMP CONTROLLED BY 2 TWO-WAY SWITCHES

AIM: To give connection to one Lamp controlled by one Switch.

MATERIAL REQUIRED:

TOOLS REQUIRED:

1. Wooden wiring board, two way switches (2 No's),
2. wooden round blocks,
3. batten lamp holders,
4. connector,
5. screw driver,
6. wires,
7. wire clips,
8. nails,
9. wood screws,
10. Poker and Bulb.

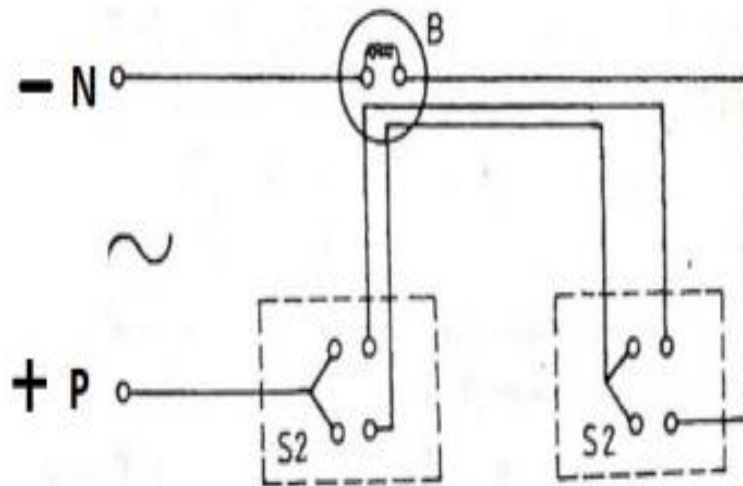
PROCEDURE:

1. The outline of the wiring diagram is marked on the wooden wiring board.
2. Clips are nailed to the board, following the wiring diagram.
3. Aluminum Sheet of (120 X 156 X 0.30 mm)
4. Wires are stretched and clamped with the clips.
5. Round blocks (3 No's) are screwed onto the board, as per the diagram
6. Wires connected to the holders and switches which are then screwed onto the round blocks. 6. Bulb is fitted to the holder.
7. The wiring connections are then tested, by giving power supply.

RESULT & CONCLUSION:

The electrical circuit, for one light controlled by two way switches is thus made.

DIAGRAM:



One Lamp controlled by two way switches

VIVA QUESTIONS:

1. What are safety precautions required in house wiring section
2. What is house wiring?
3. What Is difference between one way switch and two way switch?
4. What are safety precautions required in house wiring section?

WELDING

Introduction

Welding is metal joining process wherein localized coalescence is produced either by heating the metal to a suitable temperature, with or without the use of filler metal, with or without application of pressure. The filler material has similar composition and melting point temperature as that of the base metal. It is used to fill gap between the joint surfaces.

The welding process is divided into two main sub divisions.

Types of welding

Plastic welding

The pieces of metal to be joined are heated to the plastic state and then forced together by external pressure without the addition of filler material.

Forge welding

The work piece are placed in a forge or other appropriate furnace and heated within the area to be joined to the plastic condition. Then parts are quickly superimposed and worked into a complete union by hand or power hammering or by pressing together.

Resistance welding

In resistance welding, a heavy electric current is passed through the metals to be joined over limited area, causing them to be locally heated to plastic state and the welding is completed by the application of pressure for the prescribed period of time.

Fusion welding

In fusion welding, the metal parts to be joined are melted and then allowed to solidify pressure is not applied and filler metals may be used for this type of welding.

Gas welding:

Gas welding is a process in which the required heat to melt the surfaces is supplied by a high temperature flame obtained by a mixture of two gases. Usually the mixture of oxygen and acetylene is used for welding purpose.

Electric Arc welding

Electric arc welding is the process of joining two parts by melting their edges by an electric arc without the application of pressure and with or without use of filler metals.

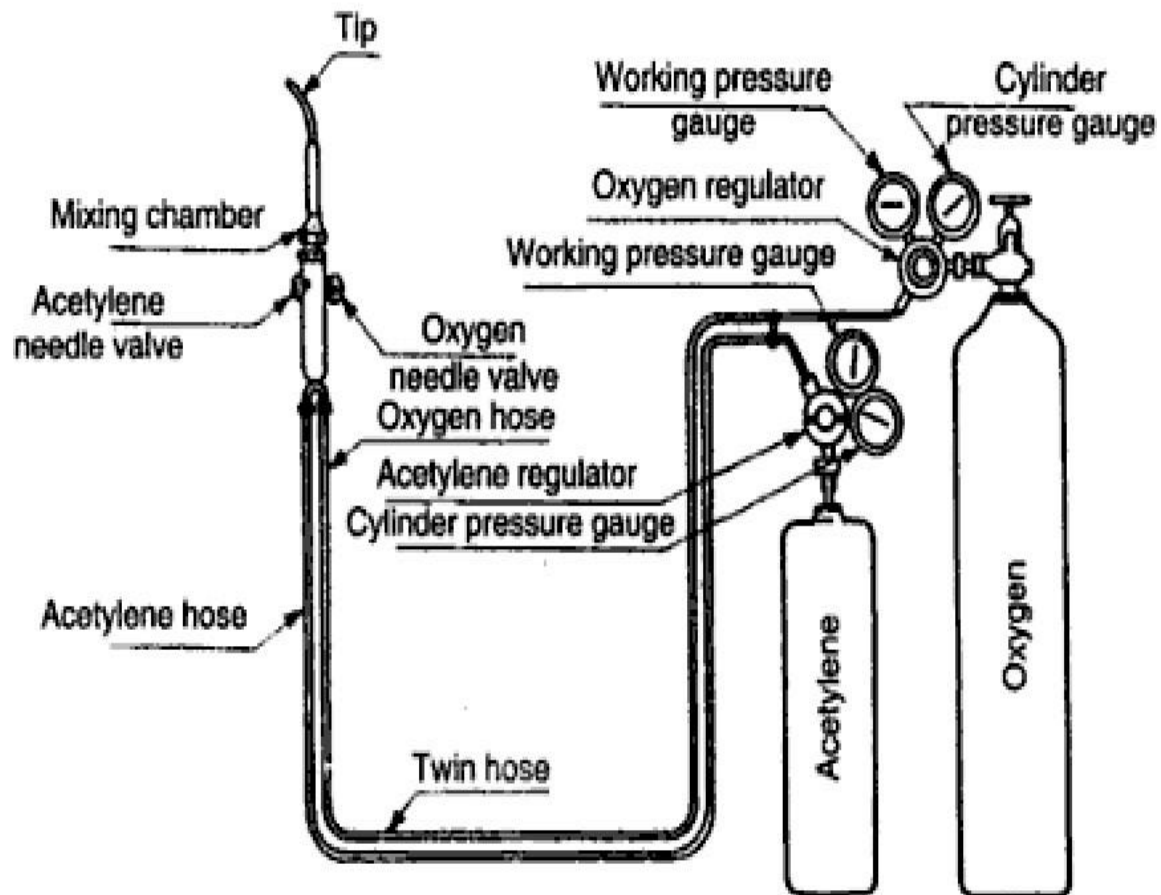
Thermit welding

Thermit welding is a fusion process in which weld is effected by pouring super heated liquid thermit steel, around the parts to be united with or without the application of pressure.

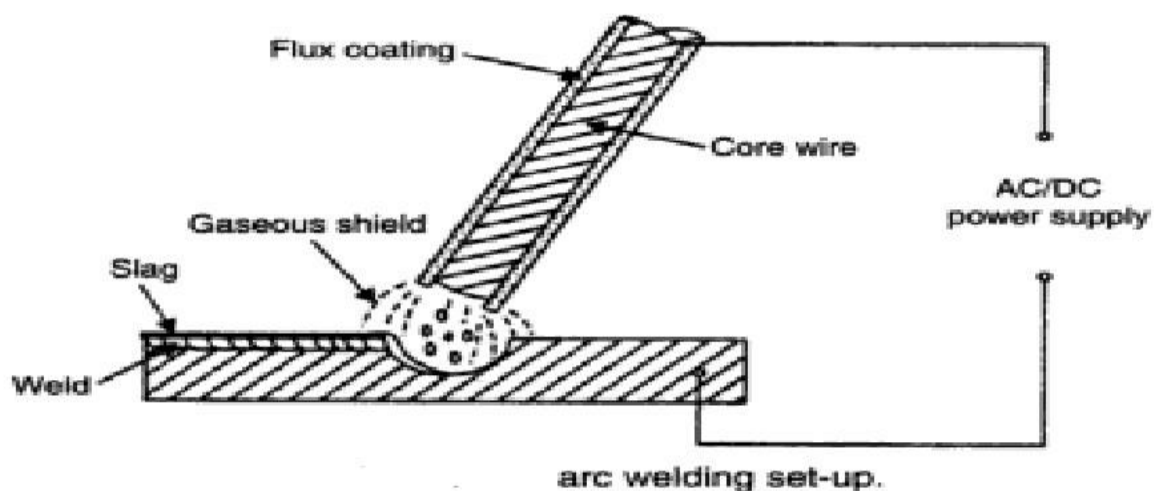
Oxy-acetylene welding

In oxy-acetylene welding oxygen and acetylene are the two gases used for producing flame. Oxygen is mainly used for supporting the combustion intensity. The oxygen and acetylene under high pressure in cylinders which are fitted with pressure regulator. Each cylinder is connected to the blowpipe by flexible hoses. The oxygen cylinders are painted black and acetylene cylinders are painted maroon. When acetylene is mixed with oxygen in correct proportions in the welding torch, ignition is takes place. The flame resulting at the tip of the torch is sufficient enough to melt and the parent material

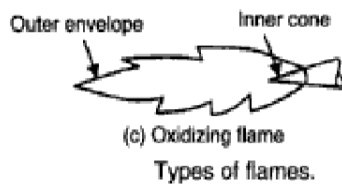
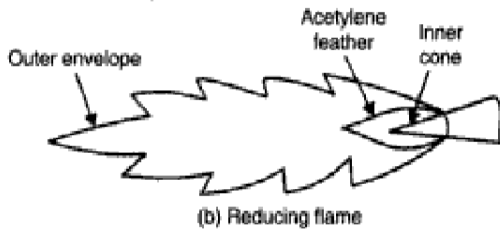
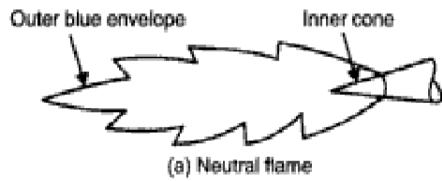
The flame temperature is about 3200oC. The filler metal rod is generally added to the molten metal pool to built up the seam for greater strength.



oxy-acetylene gas welding.



arc welding set-up.



Types of flames

1. Neutral Flame (Oxygen, Acetylene in equal proportions)
2. Oxidizing Flame (excess of Oxygen)
3. Reducing Flame (excess of Acetylene)

Neutral Flame

A neutral flame is produced when approximately equal volumes of oxygen and acetylene are mixed in the welding torch and burnt at the torch tip. The temperature of the neutral flame is of the order of about 3260°C . The flame has inner cone which is light blue in colour. The neutral flame is used for welding of mild steel, copper, aluminium, cast iron, etc.

Oxidizing Flame

If the volume of the oxygen supplied to the neutral flame is increased, then resulting flame will be oxidizing flame. The temperature of oxidizing flame is of the order of about 3482°C . Normally the outer flame envelope is much shorter. It has very small white inner cone. This flame is used to weld copper-base metals, zinc-base metals.

Reducing Flame (Carburizing Flame)

If the volume of the oxygen supplied to the neutral flame is reduced, the resulting flame will be a reducing flame. In this case, flame is recognized by acetylene feather which exists between the inner cone and outer envelope. The outer flame envelope is longer than that of the neutral flame and is usually much brighter in colour. It has an appropriate temperature of 3038°C . In this type, flames are used to weld the high-carbon steel, non-ferrous alloys, zinc-bearing alloys. Flame Adjustment Neutral flame to get the neutral flame, add sufficient oxygen to make the white cone clear and round. During the neutral flame, the gas mixture from the blow pipe consists of oxygen and acetylene in the ratio of 1:1:1 Oxidizing flame To get the oxidizing flames add, more oxygen. The white cone will become short and sharp. The flame will produce a hissing sound and will have a short length.

Carburizing flame

To get the carburizing flame adds more acetylene to the neutral flame. The white cone will become long surrounded by a feather-like portion. Flame will burn quietly and have more length. Filler metal It is the material that is added to the weld pool to assist in filling the gap. Filler metal forms an integral part of the weld. The filler metal is usually available in rod form. The rods are called filler rods.

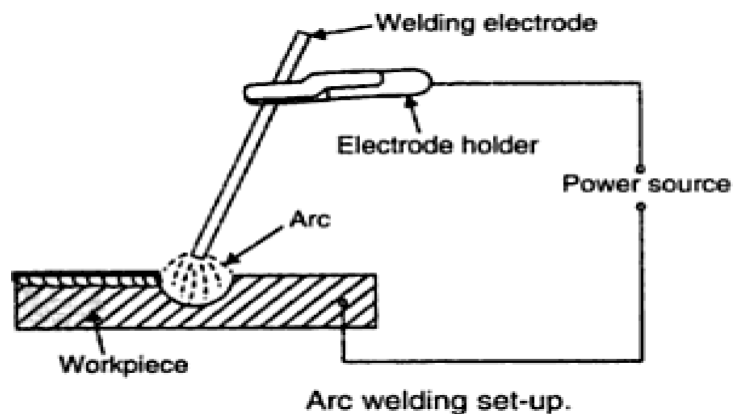
Fluxes

During the welding, if the metal is heated in the air, oxygen in the air combines with the metal to form oxides which result in poor quality, low strength welds or in some cases may even make welding impossible. In order to avoid this problem, a flux is added during the welding. The flux prevents oxidation by preventing oxygen from contacting the weld zone. No flux is used in the gas welding of steel. The most commonly used flux materials are boric acid, soda ash and sodium chloride.

Arc welding

In the arc welding process, the source of heat is electricity. In arc welding process, coalescence is produced by heating the work piece with an electric arc struck between electrode and the work piece. Welding may be carried out in air or in an inert atm. Filler material may or may not be used. The temperature of the arc is of the order of 3600°C.

KESHAV MEMORIAL INSTITUTE OF



Principle of arc welding Principle of operation

The heat required for joining the metals is obtained from an electric arc. The electric motor generator or transformer sets are used to supply high electric current and the electrodes are used to produce the necessary arc. The electrode serves as the filler rod and arc melts the surfaces so that the metals to be joined are fused together. The transformer type welding machine produces AC current. It takes power directly.

Comparison of A.C and D.C arc welding equipments :

<u>Alternating current (AC)</u>	<u>Direct current (DC)</u>
1. It is not suitable for joining non-ferrous metals	It is not suitable for both ferrous and Non-ferrous metals.
2. Only coated electrodes can be used.	Bare electrodes can also be used.
3. Consumes less power.	Consumes more power.
4. High efficiency.	Low efficiency.
5. Cost of equipment is less.	Cost of equipment is more.
6. Noiseless operation.	It gives noise.
7. Not suitable for welding thin sections.	Suitable for welding thin sections.

Electrodes

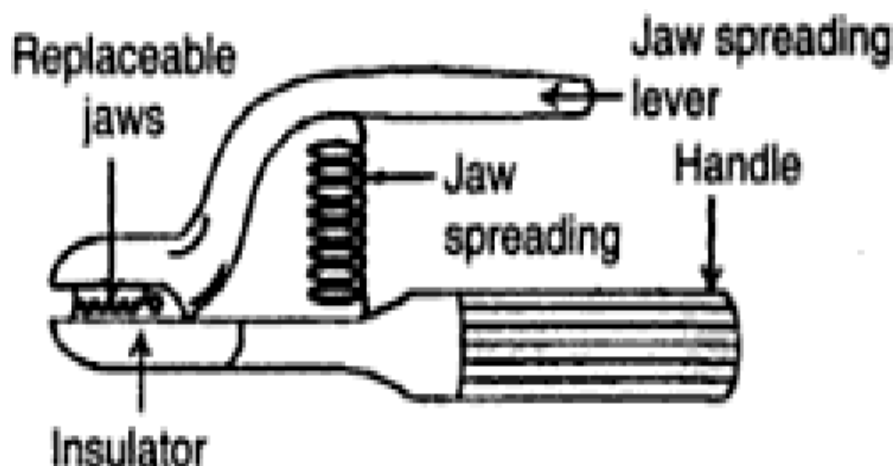
Filler rods are used in arc welding are called as electrodes. The Electrodes are made of metallic wire called core wire. It is uniformly with a protective coating called flux while fluxing an electrode about 20 mm of length is left bare at one end for holding it using electrode holder. It is used to transmit full current from electrode holder to the front end of the electrode coating.

Electrodes for Arc welding

Consumable Electrode: These electrodes are made up of copper, steel, brass etc.

Non-consumable electrode: Electrodes made up of carbon and graphite varieties are used in DC arc welding. Tungsten electrodes used for both AC and DC welding.

Electrode Holder: It is a device used for mechanically holding the electrode and conducting current to it. Electrode holder should be light, to minimize fatigue incurred by the welder. Jaws are made to hold the bare end of the electrode in either at vertical or at an angular position.



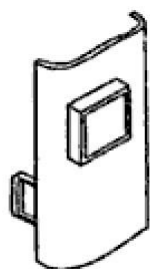
Electrode holder.

It is a device used for mechanically holding the electrode and conducting current to it. Electrode holder should be light, to minimize fatigue incurred by the welder. Jaws are made to hold the bare end of the electrode in either at vertical or at an angular position.

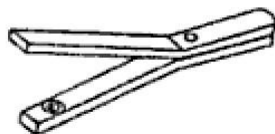
Welding Cables

Two cables are needed for welding purpose. One is used to connect the power source to electrode; another cable is connected to ground. The cables are well isolated with rubber.

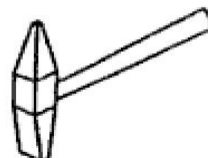
Welding Bead cleaning accessories



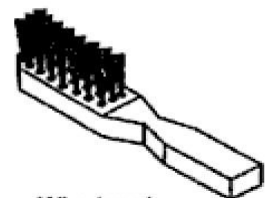
Face shield



Ground clamp



Chipping hammer



Wire brush

Chipping hammer

A chipping hammer is chisel-shaped one and it is used to remove the slag from the weld bead.

Wire Brush

A wire brush made up of stiff steel wire, embedded in wood, removes small particles of slag from the weld bead after the chipping hammer is used.

Hand Screen

It is a protective device used in arc welding. A hand shield is held in the hand of the welder and it is fitted with a suitable filter lens.

Helmet

It is used for shielding and protecting the face and neck of the welder and it is fitted with a suitable filter lens.

Tongs

Tongs are used to handle the hot metal-welding job while cleaning; they are also used to hold the metal for hammering.

Goggles

Chipping goggle is used to protect the eyes while chipping the slag. They are fitted with a plain glass to see the area to be cleaned.

Hand Gloves

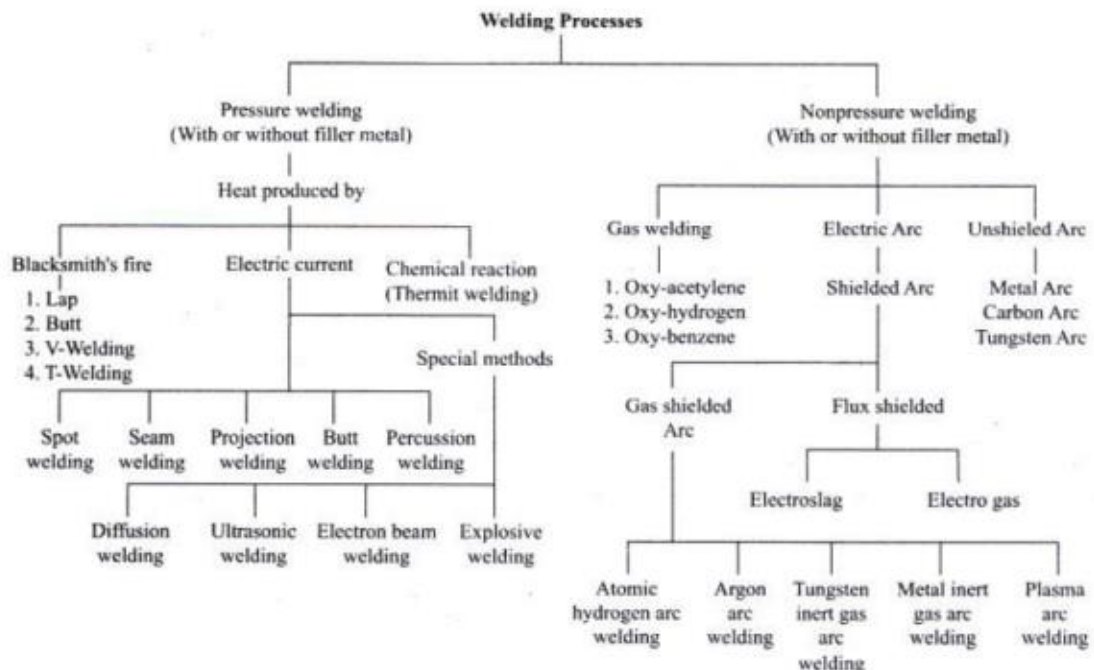
Hand gloves are used to protect the hands from electrical shock, arc radiation and hot *spatters*.

Advantages of arc welding

- * Flux shielded manual metal arc welding is the simplest of all the arc welding processes.
- * The equipment can be portable and the cost is fairly low.
- * This process finds innumerable applications, because of the availability of a wide variety of electrodes. A big range of metals and their alloys can be welded.

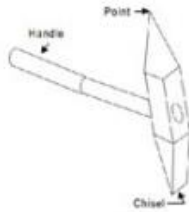
Disadvantages of arc welding

- * Because of the limited length of each electrode and brittle flux coating on it, Mechanization is difficult.
- * In welding long joints, as one electrode finishes, the weld is to be progressed with the next electrode. Unless properly cared, a defect may occur at the place where welding is restarted with the new electrode.
- * It cannot be used to weld metal thickness less than 1.6 mm.





Earth Clamp



Chipping Hammer



Steel Wire Brush



Welding Face Shield /
Hand Shield



Welding Helmet / Head Shield



Iron Tong



Leather Hand Gloves

EXERCISE NO: 1

LAP JOINT

Aim : To join the given two work pieces as a lap joint by arc welding.

Material used

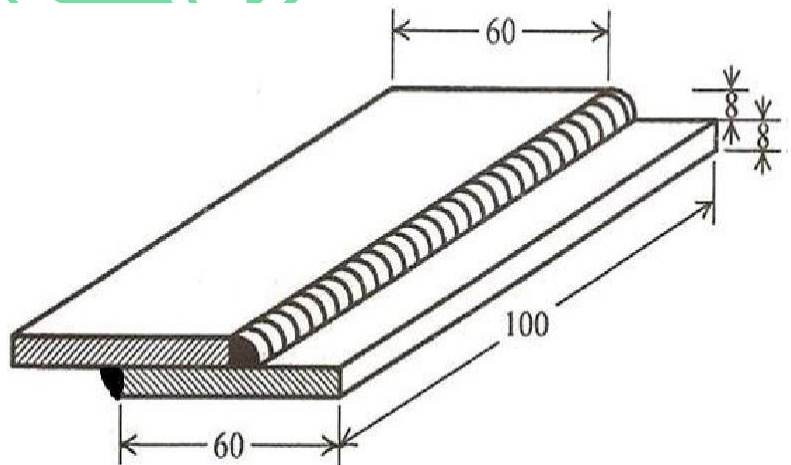
Mild Steel plates.

Tools required

- Welding power supply
- Welding rod
- Electrode holder
- Gloves and apron
- Shield and goggles
- Flat file
- Chipping hammer
- Wire brush AND Earthing clamps

Procedure

1. First of all, the work pieces must be thoroughly cleaned to remove rust, scale and other foreign materials.
2. Then the given work pieces are placed on the table in such a way that one work piece is placed on the other work piece like the LAP joint is formed.



3. Appropriate power supply should be given to the electrode and the work pieces.
4. Now the welding current output may be adjusted.
5. When current is passed, arc is produced between the electrode and work pieces.

6. Then the welding is carried out throughout the length.
7. As soon as the welding process is finished, switch off the current supply and allow the work piece to cool.
8. Slag is removed by chipping process with the help of chipping hammer.
9. Finally using wire brush, welded portions are cleaned.

Result : Thus the given two work pieces are joined as a lap joint by arc welding.



Exercise No:2

Single V butt joint

Aim

To join the given two work pieces as a single 'v' butt joint by arc welding.

Material used

Mild Steel plates.

Tools required

- Welding power supply
- Welding rod
- Electrode holder
- Gloves and apron
- Shield and goggles
- Flat file
- Chipping hammer
- Wire brush
- Earthing clamps

Procedure

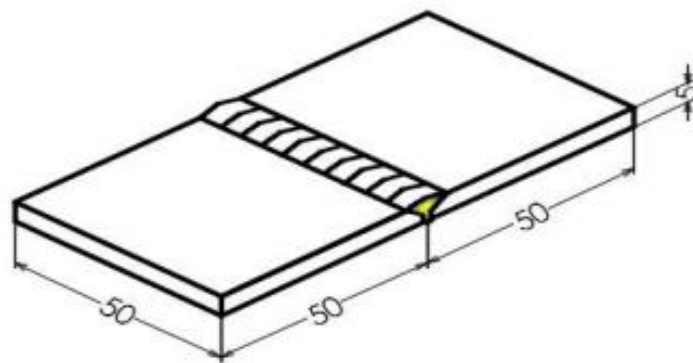
1. First of all, the work pieces must be thoroughly cleaned to remove rust, scale and other foreign materials.
2. Then the given work pieces are placed on the table in such a way that work pieces are brought close to close to each other so that it forms a V shapes when the plates butt each other.
3. Appropriate power supply should be given to the electrode and the work pieces.
4. Now the welding current output may be adjusted.
5. When current is passed, arc is produced between the electrode and work pieces.
6. Now set the two work pieces in correct position and maintain the gap 3mm and tag at both ends of the work pieces as shown in figure.
7. Then the welding is carried out throughout the length.

8. As soon as the welding process is finished, switch off the current supply and allow the work piece to cool.
9. Slag is removed by chipping process with the help of chipping hammer.
10. Finally using wire brush, welded portions are cleaned.

Result

Thus the required 'single V butt joint' is made by arc welding process.

DIAGRAM:



All dimensions are in mm

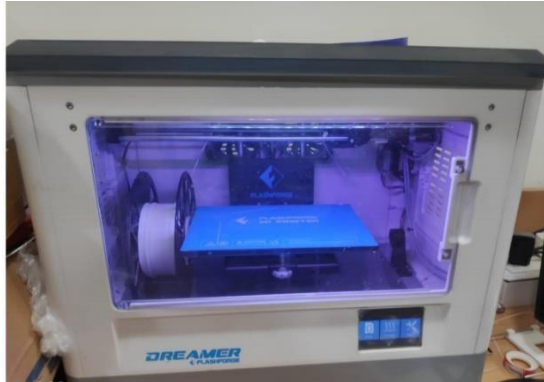
Viva Questions

Trade: Welding

01. What is meant by welding?
02. What is the name of item coated over a welding rod?
03. What is the name of tool to remove the slag from welded portion?
04. What is meant by Butt welding?
05. . What is the advantage of 'v' Butt joint?
06. Name the main tools used in the welding shop?
- 07.. What type of transformer is used in welding shop?
08. What are the safety precautions to be followed in welding shop?

3D PRINTER

The action or process of making a physical object from a three-dimensional digital model, typically by laying down many thin layers of a material in succession.



Introduction

3D Printing is a process for making a physical object from a three-dimensional digital model, typically by laying down many successive thin layers of a material. It brings a digital object (its CAD representation) into its physical form by adding layer by layer of materials.

The first documented iterations of 3D printing can be traced back to the early 1980s in Japan. In 1981, Hideo Kodama was trying to find a way to develop a rapid prototyping system. He came up with a layer-by-layer approach for manufacturing, using a photosensitive resin that was polymerized by UV light.

3D printing provides many advantages like few are simplicity, reliability and precision etc. This makes it one of the most widely used for making components which can be used as concept components. 3D printing is the most widely used additive manufacturing processes in the current industry not only limited to engineering

With this technology, objects can be built with production-grade thermoplastics¹. Objects are built by heating a thermoplastic filament to its melting point and extruding the thermoplastic layer by layer.

Special techniques can be used to create complex structures.

Advantages of 3D PRINTING:

These advantages include those related to design, time and cost, amongst others.

- Flexible Design. ...
- Rapid Prototyping. ...
- Print on Demand. ...
- Strong and Lightweight Parts. ...

- Fast Design and Production. ...
- Minimizing Waste. ...
- Cost Effective. ...
- Ease of Access.

Material Used in 3D Printing:

Plastic is still the most popular material used for 3D printing. As the 3D-printing market value increases, the list of what materials can be used also grows. Raw materials such as metal, graphite, and carbon fibre are commonly used for 3D printing, though at-home use is mostly limited to PLA for now.

Problems that solves by 3D Printing:

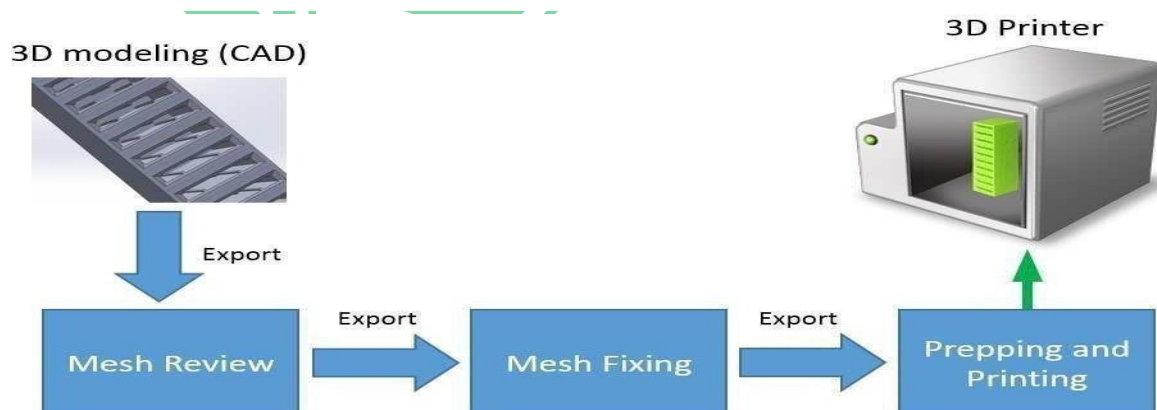
The following Top Engineering Problems can be solved with 3D Printing

- 1) Speed and lead time. Quality manufacturing takes time. ...
- 2) Cost reduction. ...
- 3) Risk mitigation. ...
- 4) Design flexibility. ...
- 5) Materials & sustainability.

3D Printing ways in academics:

There is an array of uses for 3D printing across academic disciplines: biology students can print organs, chemistry students can study 3D printed molecules, graphic design students can create 3D versions of their art and history students can print historical artefacts

Flow diagram of 3d printing process:



Components of 3D Printer: -

1. Axes

Fixed Rods The three axes that the 3D printer utilizes are on the Cartesian coordinate system. The linear fixed rods are maintained at right angles to each other and each represents a coordinate axis.

Movement The timing belts and pulleys allow the movement of the hot end (or the print bed, depending on the type of 3D printer) along each axes according to the g- code (generated by slicing software). The stepper motors power this movement.

2. Extruder

Extrusion is the feeding of filament into the hot end of the 3D printer. This movement is also powered by a stepper motor.

Retraction This mechanism is the pulling of the melted filament from the hot end. This movement is primarily programmed through the g-code to prevent the formation of unwanted filament creating a bridge between two areas. The bridging of unwanted filament is referred to as stringing or the formation of cobwebs.

Dual Extrusion Some models of 3D printers are equipped with dual extrusion capabilities. This allows for mixed material objects to be printed. Dual extrusion can be used to print out complex objects with a different colour material as the support, making it easy to differentiate between the object and the support.

3. Hot End

The hot end is heated to temperatures ranging from 160 C to 250 C, depending on the type of filament to be used. The hot end melts the filament and pushes the melted filament through the nozzle. The hot end needs to be thermally insulated from the other components of the 3D printer to prevent any damage.

4. Print Bed

Heated Print beds that are heated improve print quality of 3D printed objects. The heated bed is heated to the glass transition temperature of the filament being used. This allows the model layers to slightly melt and stick to the heated bed.

Non-Heated Print beds that are not heated require adhesion in the form of glue, tape, hairspray, etc. In the innovation lab, painters tape is frequently used for adhesion.

5. Filament

Filament is a consumable used by the 3D printer to print layers. Filament comes in a variety of materials and colors. Filament can be composed of metal, wood, clay, biomaterials, carbon fibre, etc.

PLA: - PLA is a thermoplastic that needs to be heated to temperatures from 160C to 220C. PLA is also biodegradable and emits slight odours. PLA is most frequently used in the Innovation Lab on all 3D printers.

Preparing your 3D Model for print in AutoCAD software:-

These are following step for 3D printing of model

1. Install the 3D print software
2. Check repair option in this software
3. Set the nozzle parameter and build tack temperature according to the printer guide.

Step:-1 Prepare the design Model using Designing Software

AutoCAD Step:-2 Convert the designed Model file into .Stl format.

Step:-3 Prepare the design model for printing using relevant Software then set all parameter (nozzle temp., bed temp, etc) and also repair your design using software. Then load the file in .stl format to send to 3d printing machine

Step:-4 ON the 3D Printer and load the filament in nozzle and give the command print.

Precautions of 3D Printer machine: -

The following precautions are to be taken before printing the design in 3D Printer

1. Mechanical: Do not place limbs inside the build area while the nozzle is in motion. The printer nozzle moves in order to create the object.
2. High Temperature: Do not touch the printer nozzle – it is heated to a high temperature in order to melt the build material.
3. Always buy replacement parts from the manufacturer for safety related equipment
4. Choose an area that has adequate ventilation and exhaust capability.

Experiment No -1

HEXAGONAL PYRAMID

Aim: To Print/make a Hexagonal Pyramid with given dimensions using 3D Printer

Material Required: PLA (Poly Lactic Acid) of 1.75mm diameter

Machines Required

1. Computer System with AutoCAD software installed.
2. 3-D Printing machine with PLA filament loaded.

Procedure

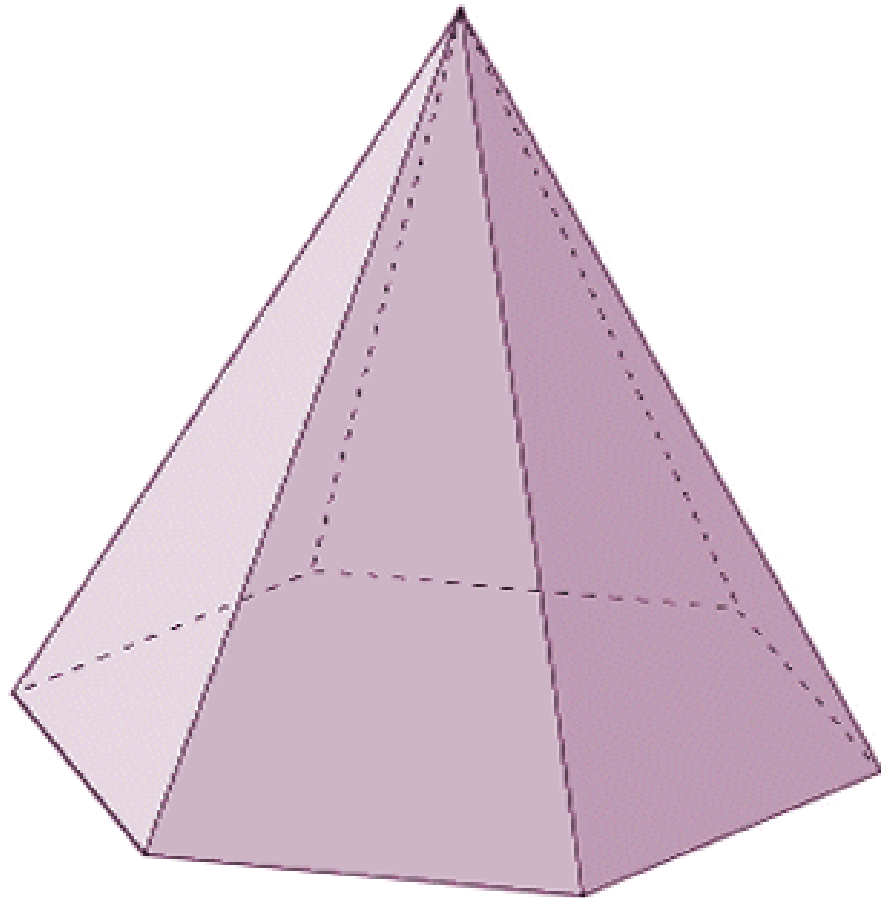
1. Turn on the computer.
2. Open AutoCAD software and set the units as millimeters.
3. Prepare the Model of hexagonal Pyramid using appropriate commands based on geometry of the model in AutoCAD software.
4. Export the file from **.dwg** format to **.stl** format by giving a specific name and choose the location to save it in the computer.
5. Open this **.stl** file from the specific location saved earlier. It will show the file in 3D printer connected to the computer.
6. Watch the model in bottom View direction to know whether it is aligned with work table of 3D printer or not.
7. Proceed with next step if the base of the model color is blue. If not set the work table.
8. Go to control module and slice the model and give the nozzle with which the filament is connected and set the temperatures.
9. Then, go to local save and send the model to the printer.
10. Finally check the file name and start printing the object.

Precautions:

1. Do not place limbs inside the build area while the nozzle is in motion.
2. High Temperature: Do not touch the printer nozzle / bed / platform
3. check the platform levels before starting of printing the object
4. Choose an area with adequate ventilation and exhaust capability to place the machine.

Result:

The hexagonal Pyramid is thus printed as per given dimensions using 3D Printer



EXPERIMENT -2

CYLINDER

Aim: To Print/make a Cylinder with given dimensions using 3D

Printer **Material Required:** PLA (Poly Lactic Acid) of 1.75mm

diameter **Machines Required**

1. Computer System with AutoCAD software installed.
2. 3-D Printing machine with PLA filament loaded.

Procedure

1. Turn on the computer.
2. Open AutoCAD software and set the units as millimeters.
3. Prepare the Model of hexagonal Pyramid using appropriate commands based on geometry of the model in AutoCAD software.
4. Export the file from **.dwg** format to **.stl** format by giving a specific name and choose the location to save it in the computer.
5. Open this **.stl** file from the specific location saved earlier. It will show the file in 3D printer connected to the computer.
6. Watch the model in bottom View direction to know whether it is aligned with worktable of 3D printer or not.
7. Proceed with next step if the base of the model color is blue. If not set the worktable.
8. Go to control module, slice the model, and give the nozzle with which the filament is connected and set the temperatures.
9. Then, go to local save and send the model to the printer.
10. Finally check the file name and start printing the object.

Precautions:

1. Do not place limbs inside the build area while the nozzle is in motion.
2. High Temperature: Do not touch the printer nozzle.
3. Always buy spare parts from the manufacturer for safety of the machine.
- 4.

Choose an area with adequate ventilation and exhaust capability to place the machine

5. .

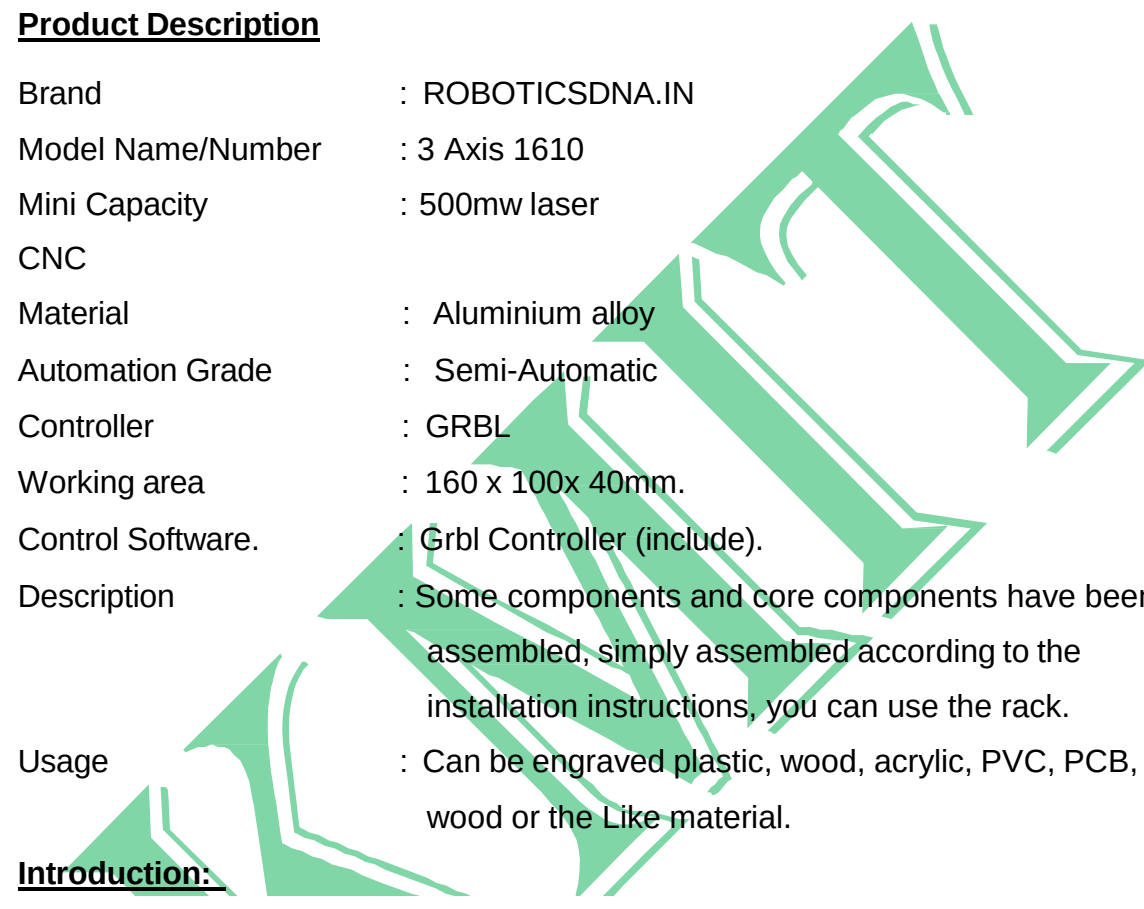


shutterstock.com • 2081471311

CNC MACHINE

A computer numerical control (CNC) Engraving machine is a computer-controlled cutting machine, which typically mounts a hand-held router as a spindle used for cutting wood/acrylic material. It works on the Cartesian coordinate system (X, Y, Z) for 3D motion control.

Product Description



Brand	: ROBOTICS DNA.IN
Model Name/Number	: 3 Axis 1610
Mini Capacity	: 500mw laser
CNC	
Material	: Aluminium alloy
Automation Grade	: Semi-Automatic
Controller	: GRBL
Working area	: 160 x 100x 40mm.
Control Software.	: Grbl Controller (include).
Description	: Some components and core components have been assembled, simply assembled according to the installation instructions, you can use the rack.
Usage	: Can be engraved plastic, wood, acrylic, PVC, PCB, wood or the Like material.

Introduction:

The CNC carving technology is produced by the combination of traditional carving technology and modern CNC technology. And the application of CNC Engraving Machine is very wide. The CNC carving technology covers a variety of advanced technologies, including computer-aided design (CAD) technology, computer-aided manufacturing (CAM) technology, numerical control (NC) technology, precision manufacturing technology. It is a new and unique processing CNC engraving machine technology.

Compared with traditional processing technology, CNC engraving machine has the advantages of high processing material efficiency, high speed, good surface quality, and high- cost performance. The traditional engraving process is difficult to engrave complex work pieces, and it is also time-consuming.

Structure of CNC Engraving Machine:

The electric control cabinet, the control computer, and the engraving CNC controller constitute the CNC engraving machine. The CNC engraving machine is the structure of the machine, composed of a workbench base, a fixed workbench, a movable gantry and a spindle. The spindle cutter moves in the X and Z-axis directions, while the gantry moves in the Y-axis direction. This constitutes a three-axis and three-axis linkage structure. In terms of function, the CNC engraving system consists of two parts.

The function of CNC engraving software CAD/CAM is to design graphics and text, shape and output processing path data of engraved products. The output data is a G code that can be recognized and interpreted by the CNC engraving machine. In this way, the CNC engraving machine can smoothly realize the processing.

That is the CNC engraving machine mentioned above. The engraving machine executes the G code output by the CAD/CAM software, completes various complex processing actions, makes the X-axis, Y-axis, and Z-axis move accurately, and finally completes the engraving of the work piece smoothly.

Operation principle of CNC engraving machine processing :

The commonly used graphic design software for CNC engraving machine is CAD design. It introduces three details of engraving graphics processing, path programming and processing operations.

Engraving Process Technology:

Edit the path of the CNC engraving machine.

The CNC engraving machine relies on the computer to control the machine's execution parts to complete the processing work. Mechanical engineers use design software such as Vectric Aspire to draw accurate engraving graphics from the designed engraving drawings. The engraving processing graphics is the basis for controlling the command generation of the engraving machine. Therefore, mastering the engraving machine drawing software's marking commands is a prerequisite for smooth engraving and processing graphics.

The machine tool path is a processing program used to control how the CNC engraving machine uses the knife, determines the cutting amount and the way of the knife during the machining process. As mentioned above, the machine tool path is calculated by the CNC engraving software based on the design data graphics. In layman's terms, the tool path can control the tool centre's running track during the machining process according to the engraving instructions. Therefore, the key to the machine tool path lies in the tool path's correct calculation, according to the engraving software's planning.

Steps to generate a path:

The following are the six major steps to generate a path:

- Choose engraving processing graphics.
- Choose the engraving processing method and determine the engraving range.
Select the engraving tool and modify the calculation settings.
- Choose the engraving material and set the cutting amount.
- Generate engraving processing parameters.
- Generate machine tool path.

Choose a tool for engraving:



Before the CNC engraving machine is started, a person needs to perform operations such as loading processing programs, selecting loading and unloading tools, clamping processing materials, and setting cutting parameters. The CNC engraving machine then performs cutting processing according to the machine tool path calculated by the engraving software. The processing process is completely mechanized and automated. In the process of processing, manual intervention is not required.



There is a wide range of applications which you can engrave with our machine accessories such as deep 2D and 3D engravings in stainless steel, aluminium, granite, marble and wood.

The LaserGRBL produces GRBL compatible g-code so you can use it to etch and engrave an image with your CNC router. This engraving method is great for creating customized metal gifts like metal drinks bottles, dog tags, jewellery, nameplates and locker key fobs

3-axis machining is suitable for performing drillings and threaded holes in-line with an axis or creating planar milled profiles. It is possible to machine a component from all six sides using 3-axis machining, but you will need a new fixturing setup for each side, which is an expensive process.

CNC machining is the process of removing material from a work piece until the desired shape is configured. These machines have at least a 3 axes and operate along an XYZ plane: X axis (vertical), Y axis (horizontal), and a Z axis (depth).

Applications of CNC Engraving Technology:

- a. Processing wooden door pressure line.
- b. Decorative patterns on the surface of processed wooden doors.
- c. Processing solid wood composite doors inner-frame
- d. Processing keyhole of the wooden door
- e. Machining the tongue-and-groove of double door or son-mother door
- f. Template for processing arc or special-shaped parts



(a)



(b)



(d)



(f)

Exercise 1

KMIT LOGO

AIM: To engrave the given logo on acrylic/wooden piece

Material Required: Wooden / Acrylic Sheet

Apparatus: Computer System with net connection, 3-axes CNC engraving machine along with tool kit.

Experimental procedure:

1. Turn on the computer.
2. Select an image, which has to be engraved on wood/acrylic material.
3. Convert image from **jpeg** file to **.svg** format by using online conversion software.
4. Convert **.svg** file in to **G-code** file by using **J-Cut software** and save it in **.nc or G-code** File format
5. Now save **.nc or G-code file** in a specific folder
6. Open **LGrbl or Candle software** and open **.nc file, which** is already saved.
7. Check the status of engraving machine in the app as **“idle”**
8. Adjust the control keys in the app(candle) to place the tool bit tip where the engrave is to be done on the wooden / acrylic piece.
9. Enter the values of parameters such as,
a. Spindle speed=800 b. Feed=100 c. Step = 0.01
10. Click on **send** button to start engrave process.

Precautions:

1. Always stop the machine before making adjustments.
2. Do not change the spindle speeds until the machine comes to a complete stop.
3. Always wear protective eye protection.

Result: The given logo is engraved on the Wooden / Acrylic sheet by using 3-axis CNC Engraving Machine.



Exercise 2

I Phone LOGO

AIM: To engrave the given logo on acrylic/wooden piece

Material Required: Wooden / Acrylic Sheet

Apparatus: Computer System with net connection, 3-axes CNC engraving machine along with tool kit.

Experimental procedure:

11. Turn on the computer.
12. Select an image, which has to be engraved on wood/acrylic material.
13. Convert image from **jpeg** file to **.svg** format by using online conversion software.
14. Convert **.svg** file in to **G-code** file by using **J-Cut software** and save it in **.nc or G-code** File format
15. Now save **.nc or G-code file** in a specific folder
16. Open **LGrbl or Candle software** and open **.nc file, which** is already saved.
17. Check the status of engraving machine in the app as **“idle”**
18. Adjust the control keys in the app(candle) to place the tool bit tip where the engrave is to be done on the wooden / acrylic piece.
19. Enter the values of parameters such as,
a. Spindle speed=800 b. Feed=100 c. Step = 0.01
20. Click on **send** button to start engrave process.

Precautions:

4. Always stop the machine before making adjustments.
5. Do not change the spindle speeds until the machine comes to a complete stop.
6. Always wear protective eye protection.

Result: The given logo is engraved on the Wooden / Acrylic sheet by using 3-axis CNC Engraving Machine.



 iPhone

TRADES FOR DEMONSTRATION

1. POWER TOOLS

Introduction:

Power tool is a powered by an electric motor, a compressed air motor, or a gasoline engine. Power tools are classified as either stationary or portable, where portable means handheld. They are used in industry, in construction, and around the house for cutting, shaping, drilling, sanding, painting, grinding, and polishing. Stationary power tools for metalworking are usually called Machine tools.

The lathe is the oldest power tool, being known to the ancient Egyptians. Early industrial revolution-era factories had batteries of power tools driven by belts from overhead shafts. The prime power source was a water wheel or a steam engine.

1. Power hacksaw:

A power hacksaw is a type of hacksaw that is powered either by its own electric motor (also known as electric hacksaw) or connected to a stationary engine. Most power hacksaw are stationary machines but some portable models do exist. Stationary models usually have a mechanism to lift up the saw blade on the return stroke and some have a coolant pump to prevent the saw blade from overheating.

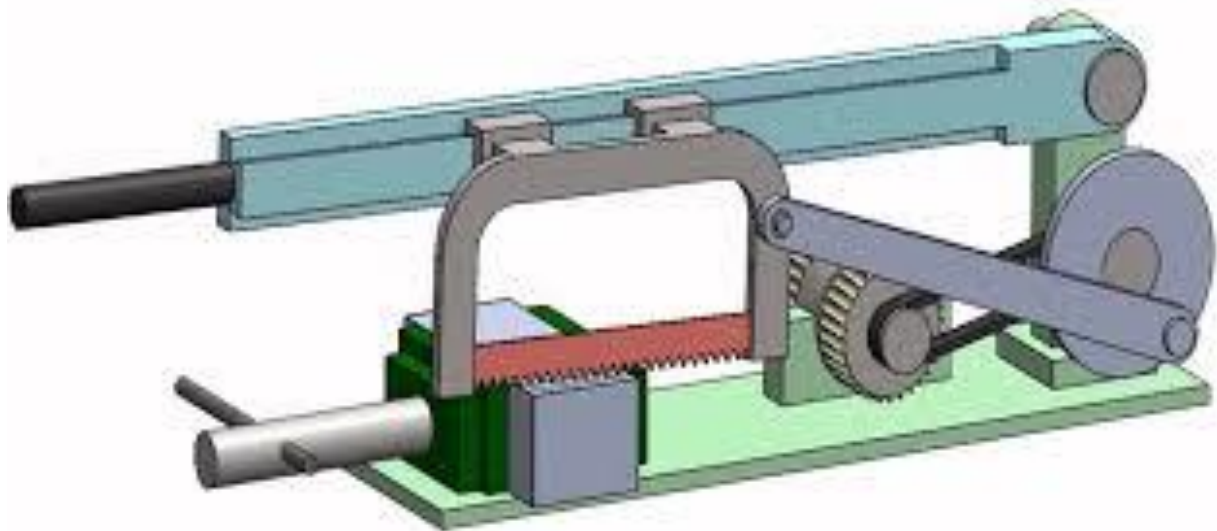


Fig. Power Hacksaw

While stationary electric hacksaw are reasonably uncommon they are still produced but saws powered by a stationary engines have gone out of fashion. The reason for using one is that they provide a cleaner cut than an angle grinder or other types of saw.

2. DRILLING MACHINE:

A drill is a tool with a rotating drill bit used for drilling holes in various materials. Drills are commonly used in woodworking, metalworking. Special designed drills are also used in medical and other applications such as in space missions.

3. BENCH GRINDER:

A bench grinder or pedestal grinder is a machine used to drive an abrasive wheel (or wheels). Depending on the grade of the grinding wheel it may be used for sharpening cutting tools such as lathe tools or drill bits. Alternatively it may be used to roughly shape metal prior to welding or fitting. A wire brush wheel or buffing wheel can be interchanged with the grinding wheels in order to clean or polish work-pieces.

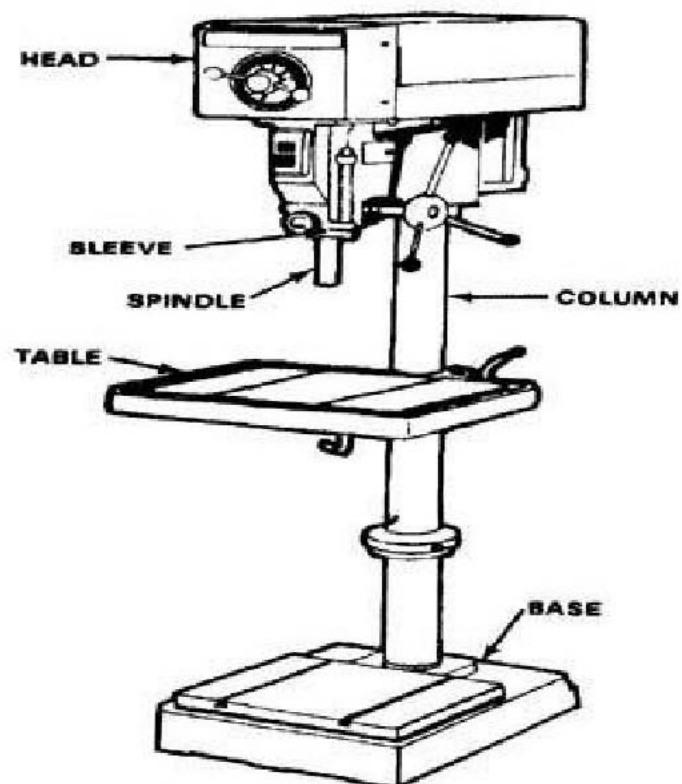
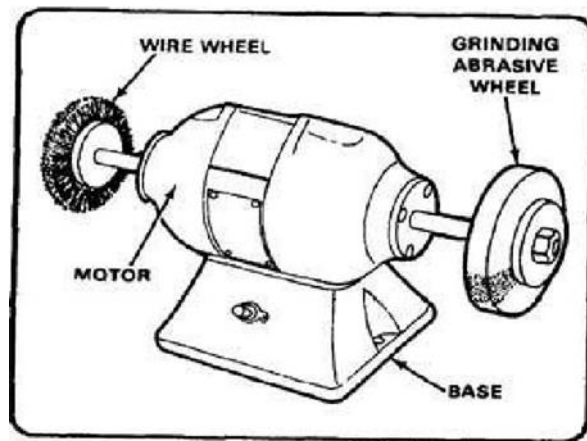


Figure 4-4. Construction of an upright drilling machine.

PLUMBING

Introduction

Plumbing deals with the laying of a pipeline. A craftsman may be perfectly proficient with the hammer, saw and other tools, but the faces difficulties with leaking pipes and overflowing toilets. Many people rush to a plumber on seeking a tripping pipe, but a person with a little knowledge of the sanitary system can control this problem easily, saving time and, one with help of few tools.

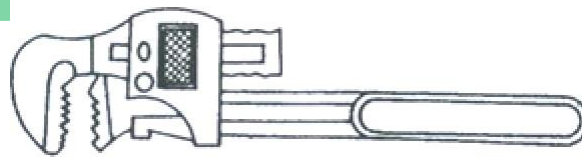
Plumbing tools

The tools used by a plumber can be classified as follows

- | | |
|----------------|--------------------|
| 1. Pipe wrench | 5. Dies |
| 2. Hacksaw | 6. Pipe cutter |
| 3. Plumb bob | 7. Files and Rasps |
| 4. Pipe vice | |

Pipe wrench

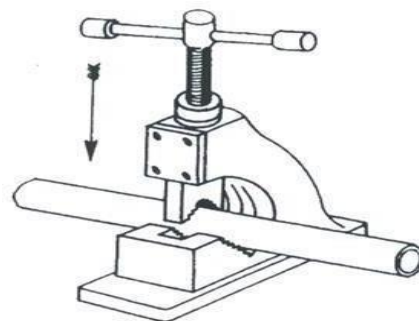
A pipe wrench is used for holding and turning the pipes, rods and machine parts. Wrenches are classified as follows. 1. Fixed wrenches 2. Adjustable wrenches.



Pipe wrench.

Pipe vice

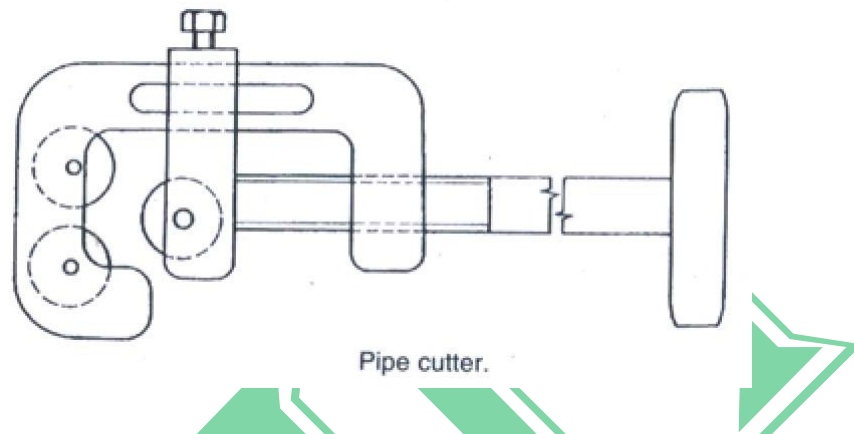
A pipe vice is fitted on the work bench. This has a set of jaws to grip the pipe and prevent it from turning while cutting, threading and fitting of bends, couplings etc. The yoke vice is commonly used in plumbing used in plumbing practice.



Pipe vice.

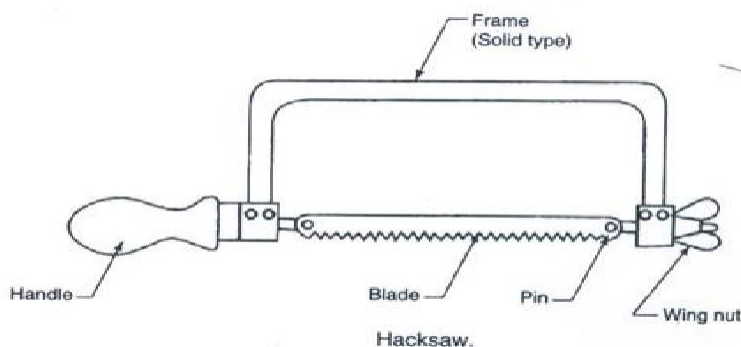
Pipe cutter

The pipe cutter mainly consists of three wheels which are hardened with sharp cutting edges along their periphery. Of these three wheels, one can be adjusted to any desired distance to accommodate different size of pipes. After adjusting the cutter on a pipe, it is around the pipe, so that the cutter wheels cut the pipe along a circle as shown in fig.



Hack saw

A hacksaw is used for cutting metal rods, bars, pipes, etc.

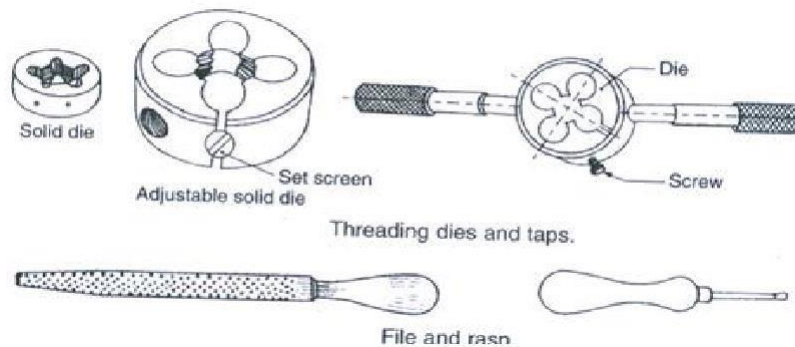


Threading dies and taps

It is used for cutting external thread on pipes. Threads are produced in various shape and sizes which are used for fitting inside a handle.

Files and rasps

The file surface is covered with sharp edged teeth and its used for removing metal by rubbing. A rasp is used for finishing the surface of the work piece.

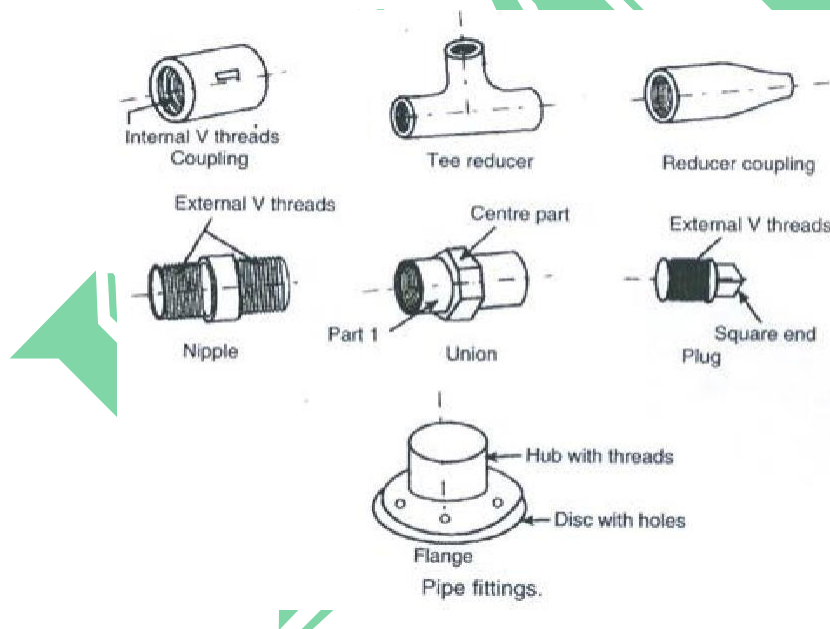


Plumb bob

It is used for check the vertical line and made up of steel or brass.

Pipe fittings

Pipe fittings are made up of wrought iron. The size of pipe fitting is designated by the size of the pipe on which it fits. some of the common pipe fittings are shown in fig.



Coupling

It is a short a cylindrical sleeve with internal threads throughout. A couplings is used for joining two pipes in a straight and bend where at least one pipe can be turned.

Union

A union is used for joining two pieces of pipes, where either can be turned. It consists of three parts, two parts joint can be screwed, in to two pipe ends, and the third on for tightening called centre part.

Nipple

A nipple is a short piece of pipe with external threads at both ends. It is used to make up the required length of a pipe line.

Elbow

An elbow is to make an angle between adjacent pipes.

Tee

A tee is a fitting that has one side outlet at a right angle to the run. It is used for a single outlet branch pipe.

Reducer

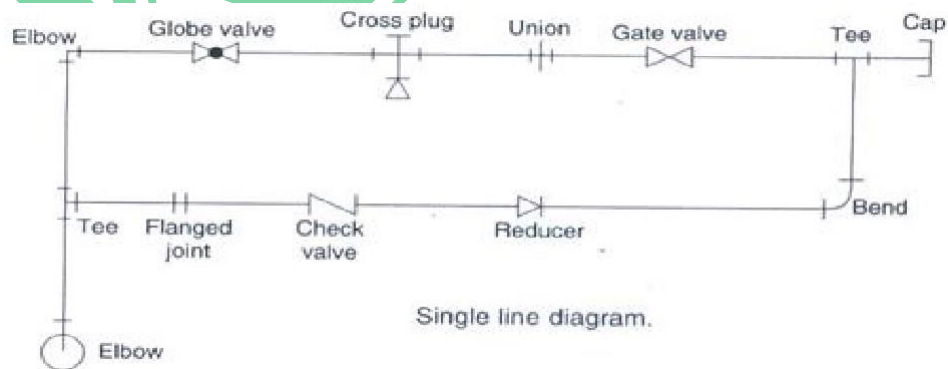
It is used to connect two different sized of pipes

Plug

It is used to screw on to a threaded opening, for closing it temporarily.

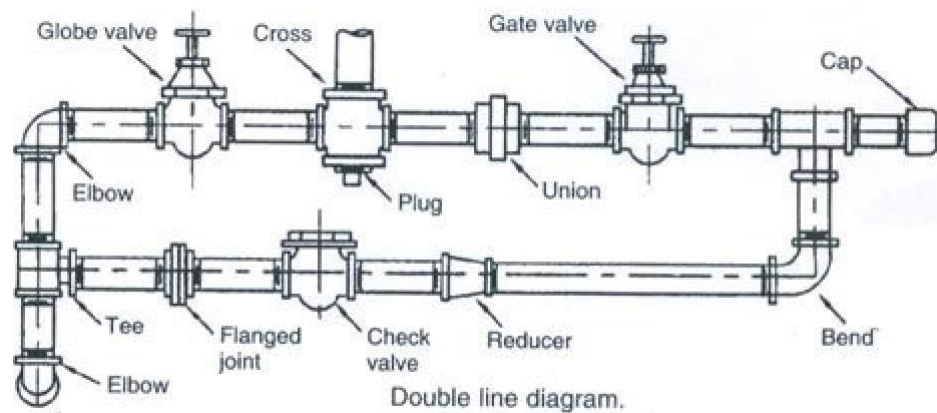
Single line diagram

Single line diagram are most commonly used in plumbing diagram. All power plants and bottling plant pipes are made by the single line piping diagram.



Double line diagram

It is used for catalogs and other applications where the visual appearance is more important.



MACHINE SHOP

Introduction

In a machine shop, metals are cut to shape on different machine tools. A lathe is used to cut and shape the metal by revolving the work against a cutting tool. The work is clamped either in a chuck, fitted on to the lathe spindle or in-between the centres. The cutting tool is fixed in a tool post, mounted on a movable carriage that is positioned on the lathe bed. The cutting tool can be fed on to the work, either lengthwise or crosswise. While turning, the chuck rotates in counter-clockwise direction, when viewed from the tail stock end.

Principal parts of a Lathe

Figure 4.1 shows a centre lathe, indicating the main parts. The name is due to the fact that work pieces are held by the centres.

Bed

It is an essential part of a lathe, which must be strong and rigid. It carries all parts of the machine and resists the cutting forces. The carriage and the tail stock move along the guide ways provided on the bed. It is usually made of cast iron.

Head stock

It contains either a cone pulley or gearings to provide the necessary range of speeds and feeds. It contains the main spindle, to which the work is held and rotated.

Tail stock

It is used to support the right hand end of a long work piece. It may be clamped in any position along the lathe bed. The tail stock spindle has an internal Morse taper to receive the dead centre that supports the work. Drills, reamers, taps may also be fitted into the spindle, for performing operations such as drilling, reaming and tapping.

Carriage or Saddle

It is used to control the movement of the cutting tool. The carriage assembly consists of the longitudinal slide, cross slide and the compound slide and apron. The cross slide moves across the length of the bed and perpendicular to the axis of the spindle. This movement is used for facing and to provide the necessary depth of cut while turning. The apron, which is bolted to the saddle, is on the front of the lathe and contains the longitudinal and cross slide controls.

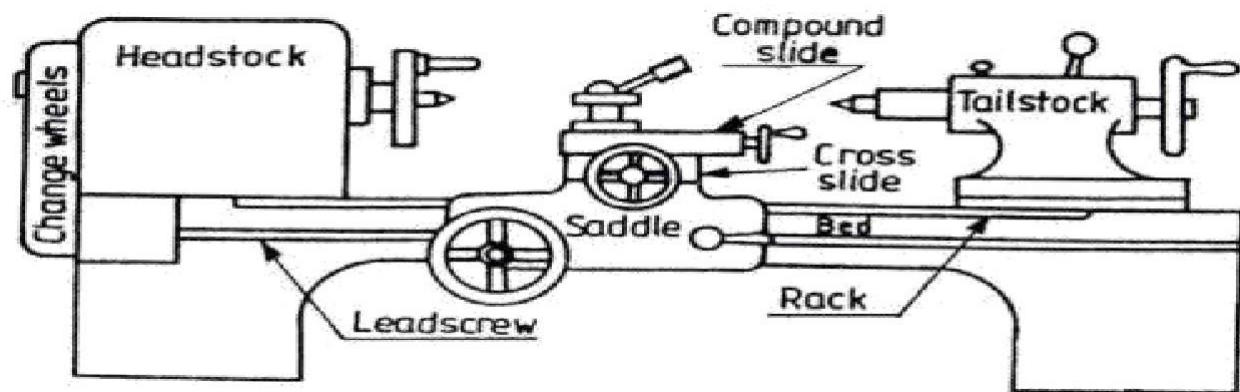


Fig :1Parts of a centre Lathe



Fig :2 three jaw and four jaw chuck

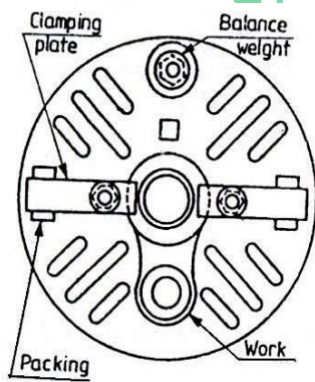


Fig:3 face plate

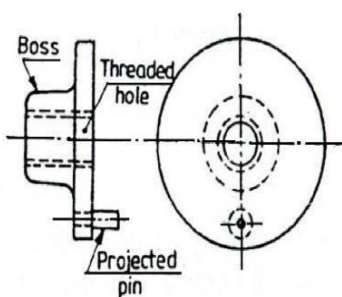


Fig :4 lathe dog and driving plate

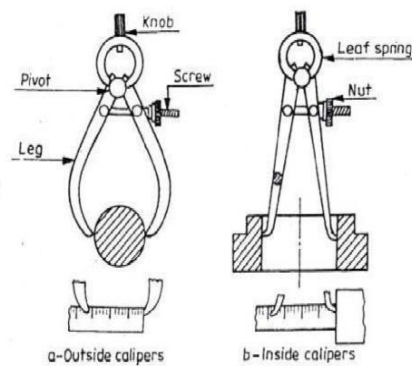


Fig: 5 calipers

Compound Rest

It supports the tool post. By swiveling the compound rest on the cross slide, short tapers may be turned to any desired angles.

Tool Post

The tool post, holds the tool holder or the tool, which may be adjusted to any working position.

Lead Screw

It is a long threaded shaft, located in front of the carriage, running from the head-stock to the tail stock. It is geared to the spindle and controls the movement of the tool, either for automatic feeding or for cutting threads.

Centers

There are two centers known as dead center and live center. The dead center is positioned in the tail stock spindle and the live center, in the head-stock spindle. While turning between centers, the dead center does not revolve with the work while the live center revolves with the work.

WORKHOLDING DEVICES

1. Three jaw chuck

It is a work holding device having three jaws (self-centering) which will close or open with respect to the chuck center or the spindle center, as shown in figure. It is used for holding regular objects like round bars, hexagonal rods, etc.

Face plate

It is a plate of large diameter, used for turning operations. Certain types of work that cannot be held in chucks are held on the face plate with the help of various accessories.

Lathe dogs and driving plate

These are used to drive a work piece that is held between centers. These are provided with an opening to receive and clamp the work piece and dog tail, the tail of the dog is carried by the pin provided in the driving plate for driving the work piece.

MEASURING INSTRUMENTS

1. outside and inside Calipers

Firm joint or spring calipers are used for transfer of dimensions with the help of a steel rule.

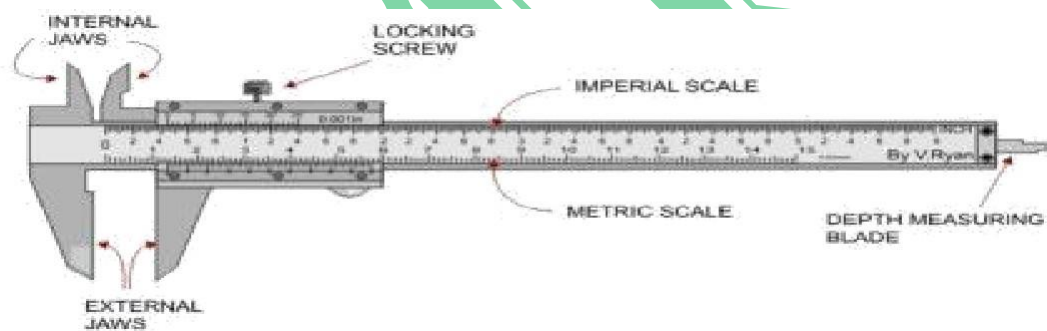
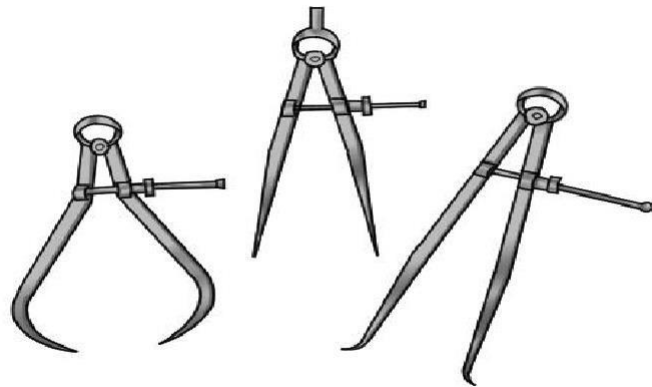


Fig: verniercaliper

2. Venire Calipers

Venire caliper is a versatile instrument with which both outside and inside measurements may be made accurately. These instruments may have provision for depth measurement also.

3. Micrometers

Outside and inside micrometers are used for measuring components where greater accuracy is required.

CUTTING PARAMETERS

1. Cutting speed

It is defined as the speed at which the material is removed and is specified in meters per minute. It depends upon the work piece material, feed, depth of cut, type of operation and so many other cutting conditions. It is calculated from the relation,

$$\text{Spindle speed (RPM)} = \frac{\text{cutting speed} \times 1000}{\pi D}$$

Where D is the work piece diameter in mm.

2. Feed

It is the distance traversed by the tool along the bed, during one revolution of the work. Its value depends upon the depth of cut and surface finish of the work desired.

3. Depth of Cut

It is the movement of the tip of the cutting tool, from the surface of the work piece and perpendicular to the lathe axis. Its value depends upon the nature of operation like rough turning or finish turning.

TOOL MATERIALS

General purpose hand cutting tools are usually made from carbon steel or tool steel. The single point lathe cutting tools are made of high speed steel (HSS). The main alloying elements in 18-4-1 HSS tools are 18 percent tungsten, 4 percent chromium and 1 percent vanadium. 5 to 10 percent cobalt is also added to improve the heat resisting properties of the tool. Carbide tipped tools fixed in tool holders, are mostly used in production shops.

TIG200A



ARGON HOSE



ARGON REGULATOR



FOOT PEDAL



HAND-HELD SHIELD,
WIRE BRUSH, ETC.

CUT40 MACHINE



CUT40 - Plasma Cutter

The Plasma Cut 40 produces a high temperature plasma stream, enabling effortless cutting of all electrically conductive materials including steel, cast Iron, stainless steel, copper, aluminium brass etc.

The Plasma Cut 40 is equipped with a high quality Italian manufactured Plasma Torch developed specifically to enhance and improve the cut quality of the Plasma Cut 40 machine. Connection of the Tecmo PCH/M 35 Plasma torch provides effortless starting of the cut with powerful, fast and accurate cutting capability, additional is the benefit of longer life cycle consumable electrodes and cutting tips. The HF contact start function allows the torch tip to be placed directly onto the work piece during start up and also during the cutting process helping to provide more accurate cutting and cleaner start of the cut. The plasma Cut 40 is an exceptional machine that is suitable for a wide range of applications including sheet metal fabrication, light industrial use, site work, automotive, ducting work, repair and maintenance services.